

Reengineering Payment System Efficiency

A Simulation-Based Comparison of Traditional, Hybrid, and Atomic Settlement Architectures

*An XRP Ledger Calibration under the Warehouse of Payments
Framework*

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This thesis was written in accordance with the academic integrity policy of Tilburg University. No generative AI tools were used in the writing of this thesis beyond standard spell-checking and grammar correction.

Abstract

This thesis investigates when blockchain-based settlement can improve efficiency and risk allocation relative to traditional Financial Market Infrastructures. A discrete-time agent-based simulation compares Real-Time Gross Settlement (RTGS), Deferred Net Settlement (DNS), and XRP Ledger atomic settlement (XRPL) within a unified Total Cost of Settlement function, with a permissioned wholesale CBDC (wCBDC) control and a hybrid RTGS–LSM benchmark, across cost, contagion, and cross-border dimensions. The total-cost ranking depends on how on-demand liquidity is costed: under gross capital DNS is cheapest and XRPL matches RTGS, whereas under a risk-adjusted volatility buffer XRPL is cheapest, so the threshold is conditional, not structural. Regardless, atomic settlement — shared by XRPL and wCBDC — delivers the lowest latency, reduces contagion by three-to-four orders of magnitude, and eliminates Herstatt risk; the wCBDC control attributes these to atomic settlement, not the public blockchain. Policy implications for central banks, overseers, and regulators are discussed.

Keywords: payment systems, Real-Time Gross Settlement, Deferred Net Settlement, XRP Ledger, wholesale CBDC, atomic settlement, settlement finality, Exposure Amplification Factor, Herstatt risk, Warehouse of Payments

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1 Introduction

1.1 Background and Motivation

The architecture of global payment systems stands at a critical juncture. Cross-border payments exceeded \$190 trillion in 2024 and are projected to surpass \$300 trillion within the next decade, yet the correspondent banking infrastructure that processes the vast majority of these transactions was designed for a world of batch processing, fixed operating hours, and bilateral trust relationships between financial institutions (Bank for International Settlements 2020). The cost of this architectural mismatch is no longer theoretical: financial institutions collectively maintain hundreds of billions of euros in pre-funded nostro and vostro accounts to support correspondent banking operations, representing idle capital that generates no return while providing settlement coverage for transactions that may take several days to reach finality (Bank for International Settlements 2020). Payment messaging systems such as SWIFT, while efficient in their core function of transmitting payment instructions, remain decoupled from the underlying settlement process, exposing participants to settlement risk during the gap between instruction and finality (Bank for International Settlements 2012).

These inefficiencies are not merely operational frictions — they are systemic vulnerabilities with documented consequences for financial stability. The collapse of Bankhaus Herstatt in 1974 gave rise to the concept of Herstatt risk: the principal risk arising from the temporal mismatch between the two legs of a foreign exchange transaction, where one party delivers its currency leg while the counterparty's delivery remains pending (Kahn and Roberds 2009). Despite decades of institutional development — including the introduction of Real-Time Gross Settlement (RTGS) systems such as TARGET2, the establishment of Continuous Linked Settlement (CLS), and the proliferation of Liquidity Saving Mechanisms — the fundamental trade-off between liquidity efficiency and settlement risk identified by Berndsen (2018) remains unresolved in traditional Financial Market Infrastructure (FMI) design. RTGS systems provide strong guarantees of settlement finality but require substantial intraday liquidity commitments from participating institutions, typically supported by collateralised

central bank credit (European Central Bank 2010). Deferred Net Settlement (DNS) systems recover liquidity efficiency through multilateral netting but reintroduce settlement delays and counterparty exposure during the clearing window (Manning et al. 2009).

The emergence of distributed ledger technology has introduced a new dimension to this long-standing debate. Blockchain-based settlement systems offer the prospect of near-instantaneous finality, continuous 24/7 operation, and programmable transaction logic without reliance on traditional intermediaries. Among these systems, the XRP Ledger has been proposed as a blockchain-based settlement infrastructure designed to facilitate fast and low-cost cross-border payments through atomic settlement and on-demand liquidity provisioning (XRP Ledger Foundation 2026). However, despite growing institutional interest in distributed ledger technology — evidenced by central bank experiments including Projects Jasper, Ubin, and Inthanon (Shabsigh et al. 2020) — the economic conditions under which blockchain-based settlement can improve efficiency and risk allocation relative to traditional Financial Market Infrastructures remain insufficiently understood. As Shabsigh et al. (2020) explicitly note, the majority of central bank DLT experiments were conducted without any formal assessment of comparative costs or economic efficiency, leaving a significant gap in the empirical literature.

1.2 Research Question and Objectives

This thesis addresses the following research question:

Under what economic conditions can blockchain-based settlement mechanisms improve efficiency and risk allocation relative to traditional clearing and settlement infrastructures?

To answer this question, the thesis develops a discrete-time agent-based simulation framework that compares three core settlement architectures — Real-Time Gross Settlement (RTGS), Deferred Net Settlement (DNS), and XRP Ledger atomic settlement (XRPL) — within a unified analytical framework, augmented by two reference configurations: a permissioned wholesale central bank digital currency (wCBDC), which isolates the atomic-settlement mechanism from the public-blockchain implementation, and a hybrid RTGS–LSM benchmark,

which occupies the interior of the latency–liquidity plane between the two traditional endpoints. The theoretical foundation of the analysis is the Warehouse of Payments framework developed by Berndsen (2018), which conceptualises Financial Market Infrastructures as inventory management systems governed by trade-offs between liquidity, time, and risk. Within this framework, the central hypothesis is that XRPL’s atomic settlement mechanism can simultaneously reduce liquidity requirements and eliminate recognition lag — a combination that static RTGS and DNS architectures cannot achieve by design.

The analysis pursues four specific empirical objectives, corresponding to four concrete outputs specified in the research design of Section 4:

1. To identify the threshold interest rate r^* and transaction volume λ^* at which XRPL’s total settlement cost falls below that of both RTGS and DNS, through sensitivity analysis over $r \in [0\%, 8\%]$ and $\lambda \in [10,000, 100,000]$ payments per day. The hypothesis is that there exists a threshold rate r^* above which XRPL’s total settlement cost falls below that of both traditional architectures; the simulation locates this threshold within the tested range.
2. To construct a comparative efficiency frontier positioning RTGS, DNS, XRPL, a hybrid RTGS–LSM benchmark, and a permissioned wholesale CBDC (wCBDC) on the latency–liquidity trade-off plane, showing whether XRPL shifts the frontier or merely relocates along it, and whether the wCBDC control — sharing atomic finality with XRPL but operating under central bank governance — reproduces the same frontier position.
3. To compute an Exposure Amplification Factor comparison across all five settlement configurations under hub-default stress, quantifying the contagion containment benefit of atomic settlement relative to both gross and net settlement architectures, and isolating whether containment is an atomic-settlement property or a public-blockchain one through the wCBDC control.
4. To identify the pool depth threshold MD^* — the minimum market liquidity depth at which XRPL on-demand liquidity provisioning becomes more expensive than maintain-

ing pre-funded nostro balances — across representative cross-border payment corridors, and to compare Herstatt risk exposure across correspondent banking, CLS, and the atomic payment-versus-payment settlement of both XRPL and wCBDC.

1.3 Contribution to the Literature

This thesis makes three distinct contributions to the literature on payment system economics. Theoretically, the study extends the Warehouse of Payments framework of Berndsen (2018) by incorporating blockchain-based settlement as a distinct operational regime. Within this extended framework, XRPL is conceptualised as a dynamic inventory mechanism where liquidity is continuously reallocated through decentralised market processes — in contrast to the static liquidity allocation of traditional infrastructures. The central theoretical question is whether atomic settlement can simultaneously reduce liquidity requirements and eliminate recognition lag, a combination that the Warehouse framework identifies as structurally impossible under RTGS and DNS design.

Methodologically, the study develops a unified simulation framework that combines all architectures within a single Total Cost of Settlement (TCS) function, enabling direct parametric comparison that has not previously been attempted in the academic literature. Three strands of related work define the boundaries of this contribution:

- First, agent-based simulation studies of RTGS systems, most notably Galbiati and Soramäki (2011), establish the methodological foundation this thesis extends, but are confined to single-architecture settings.
- Second, simulation-based efficiency frontier studies, particularly Leinonen and Soramäki (2005), quantify the latency–liquidity trade-off across traditional architectures but predate blockchain-based settlement.
- Third, institutional surveys of DLT experiments, including Shabsigh et al. (2020), explicitly identify the absence of formal comparative cost evaluation as an open research problem.

This thesis addresses all three gaps simultaneously.

From a policy perspective, the findings aim to inform institutions such as the Bank for International Settlements Committee on Payments and Market Infrastructures (CPMI) and the European Central Bank on the parametric conditions under which distributed ledger technology offers a credible alternative to traditional settlement infrastructures. By explicitly modelling contagion dynamics, settlement finality, cross-border liquidity frictions, and the distinct risk profile of blockchain-based settlement within a unified cost framework, the research provides a simulation-based foundation for evaluating the role of distributed ledger technology in the design of next-generation systemically important payment systems — while underscoring that adoption involves a reallocation of risk rather than its elimination, with governance, validator concentration, and legal harmonisation representing the primary prerequisites for institutional deployment.

2 Structure of the Thesis

The remainder of this thesis is organised as follows:

- **Section 3** reviews the relevant literature across four dimensions: the institutional framework of Financial Market Infrastructures, the economics of RTGS and DNS, settlement finality and systemic risk, and cross-border settlement frictions.
- **Section 4** presents the methodology, including the simulation framework, the settlement-architecture specifications, the efficiency frontier analysis, the stress-testing design, and the cross-border extension.
- **Section 5** reports the simulation results across all four empirical objectives.
- **Section 6** discusses the research limitations.
- **Section 7** concludes with policy implications and directions for future research.

3 Literature Review

This section reviews the relevant literature across four dimensions that collectively frame the research question. Section 3.1 presents the institutional framework of Financial Market

Infrastructures through the Warehouse of Payments metaphor. Section 3.2 reviews the economics of RTGS and Deferred Net Settlement. Section 3.3 examines settlement finality, recognition lag, and systemic risk. Section 3.4 analyses cross-border settlement frictions, with particular attention to Herstatt risk and correspondent banking inefficiencies. Section 3.5 identifies the research gap that motivates the present study.

3.1 The Institutional Framework of Financial Market Infrastructures

The theoretical foundation of this study is grounded in the Warehouse Metaphor introduced by Berndsen (2018), which conceptualises the system of Financial Market Infrastructures (FMIs) as a layered network — organised into a ground, middle, and upper floor — that settles the non-barter transactions of an economy. The ‘middle floor’ of this structure corresponds to large-value payment systems and is governed by a fundamental trade-off between liquidity need and settlement risk (Berndsen 2018). Within the same framework, Berndsen (2018) distinguishes three moments of settlement finality that mark the completion of a transfer order: entry (SF1), at which the order is legally accepted into the system; irrevocability (SF2), at which it can no longer be revoked by the sender; and unconditionality (SF3), at which it is completed without further conditions. The interval between SF1 and SF3 — which this thesis terms the *settlement gap* — is the window during which participants remain exposed to counterparty risk, and corresponds to the settlement and systemic risk that Berndsen (2018) identifies as a central concern of middle-floor design.

Building on this account, the present study reads the Warehouse through the lens of New Institutional Economics. Following North (1990) and Williamson (1985), FMIs can be understood as institutional mechanisms that reduce transaction costs and govern the exchange of value under conditions of uncertainty and asymmetric information; on this reading, liquidity functions as the ‘stock’ that participants must hold and mobilise, and settlement finality as the system’s output. The thesis accordingly investigates whether Distributed Ledger Technology (DLT), and specifically blockchain-based settlement, can shift the efficiency frontier identified in the Warehouse framework by altering the underlying trade-offs between liquidity, speed, and risk. However, recent theoretical work by Shin (2026) demonstrates

that blockchain decentralisation imposes structural coordination costs on validators, leading to network fragmentation that undermines monetary network effects. This tension between decentralisation benefits and fragmentation costs frames the simulation analysis of Section 4.

3.2 The Economics of RTGS and Deferred Net Settlement

The evolution of payment systems from Deferred Net Settlement (DNS) to Real-Time Gross Settlement (RTGS) has been extensively documented in the literature (Bech and Garratt 2003; Manning et al. 2009). RTGS systems such as TARGET2 deliver immediate finality and eliminate bilateral credit risk between participants, but at the cost of significantly higher intraday liquidity requirements, typically supported by collateralised central bank credit (European Central Bank 2010). The European Central Bank's TARGET2 system processes approximately 350,000 payments per day across approximately 1,000 direct participants, with peak intraday liquidity reaching up to three times average daily levels (Deutsche Bundesbank 2023). The collateral haircut applied to intraday credit — 2% in the TARGET2 framework (European Central Bank 2010) — generates an additional liquidity cost that compounds with the opportunity cost of prefunded balances.

DNS systems recover liquidity efficiency through multilateral netting, but reintroduce settlement delays and counterparty exposure during the clearing window. The BoF-PSS2 simulation framework documented by Leinonen and Soramäki (2003) establishes that well-functioning DNS systems achieve Liquidity Saving Ratios (LSR) of approximately 82%, meaning that only 18% of gross payment values need to be funded in net settlement. This efficiency gain comes at the cost of the SF1–SF3 settlement gap: under a 2-hour clearing cycle consistent with the EURO1 operating schedule (Manning et al. 2009), participants accumulate net positions over each cycle, creating counterparty exposure that must be covered by safety stock with a credit risk premium.

Liquidity Saving Mechanisms (LSMs) represent the primary institutional response to the latency-liquidity trade-off within traditional RTGS environments (Manning et al. 2009). Two core mechanisms are employed: bilateral offsetting, whereby two queued payments between the same pair of institutions are matched and settled simultaneously without additional

liquidity, and multilateral offsetting, whereby a cycle of payments across three or more institutions is resolved simultaneously, releasing all queued obligations at once. By enabling such payment matching, LSMs partially recover the liquidity efficiency of DNS without sacrificing the gross settlement finality of RTGS (Manning et al. 2009). However, LSMs remain constrained by operating hours and batch processing cycles, and their effectiveness depends critically on the timing and sequencing of payment submissions by participating institutions. The strategic dimension of intraday liquidity management is formally analysed by Bech and Garratt (2003), who model the RTGS settlement process as a liquidity management game in which banks optimise their payment timing to minimise collateral costs. Their analysis demonstrates that intraday liquidity costs increase disproportionately with queuing time due to cascade amplification, motivating the convex delay cost function $\varphi(D) = \alpha D^\beta$, where $\alpha > 0$ is a scaling parameter governing the absolute magnitude of delay penalties and $\beta > 1$ captures the convexity of cost escalation with queuing time, adopted in Section 4.3 of this thesis.

The efficiency frontier framework, operationalised by Leinonen and Soramäki (2005) using Finnish payment data, positions RTGS and DNS on the two-dimensional latency–liquidity trade-off plane. Their analysis quantifies the liquidity saving achievable at each level of settlement delay across a range of hybrid architectures, establishing the empirical foundation for the efficiency frontier analysis of Section 5.2. Critically, the analysis of Leinonen and Soramäki (2005) is confined to traditional architectures and predates blockchain-based settlement by two decades, leaving open the question of whether DLT can shift rather than merely relocate along this frontier.

Symmetry of Capital Treatment

Because the cost ordering is sensitive to how each architecture’s liquidity is costed, capital is treated symmetrically across architectures and two liquidity concepts are reported throughout. Under the *gross-capital* concept, adopted as the primary specification, each architecture is charged the full opportunity cost of the capital it deploys: prefunded balances for RTGS and DNS, and the gross corridor pool $MD \times P_{\text{XRP}}$ for XRPL. This treats the XRPL pool

as fully encumbered, exactly as RTGS collateral is, and is the most conservative test of the central hypothesis. Under the *risk-adjusted* concept, reported as an alternative, the XRPL liquidity input is reduced to the volatility buffer $MD \times \sigma_{\text{XRP}} \times P_{\text{XRP}}$ on the argument that the gross pool remains tradeable and earns market making income, so only the capital at risk during the 3–5 second window is settlement-attributable (the initial-margin analogy of Section 4.2). The risk-adjusted concept is not applied asymmetrically to XRPL alone: were an equivalent recoverability adjustment applied to RTGS collateral or DNS safety stock, their costed liquidity would fall correspondingly. The thesis therefore leads with the gross-capital ordering and treats the volatility-buffer result as the upper bound of XRPL’s potential cost advantage rather than as the headline finding. Several further calibration choices are deliberately conservative *against* XRPL: the 5% volatility assumption exceeds the realised 4.46% average, the DNS credit premium uses the Basel regulatory minimum of 1.6%, and the RTGS recycling friction is applied only to the incumbent. The asymmetries that remain (static p_{rollback} , no order-book depletion under stress) favour XRPL and are confined to the limitations of Section 6.5.

3.3 Settlement Finality, Recognition Lag, and Systemic Risk

Settlement finality is the central institutional output of any payment system. As established in Section 3.1, Berndsen (2018) distinguishes three stages of finality — entry (SF1), irrevocability (SF2), and unconditionality (SF3) — where the time elapsed between SF1 and SF3 constitutes the settlement gap. In traditional systems, this gap creates systemic vulnerabilities through the mechanism of recognition lag: the time between a participant’s insolvency and the system operator’s awareness. During this period, transactions may continue to be processed, leading to potential inconsistencies and the need to unwind previously settled positions.

The systemic consequences of recognition lag have been empirically documented in historical episodes, most notably the collapse of Bankhaus Herstatt in 1974, which demonstrated how settlement disruptions propagate across financial networks, amplifying losses and destabilising the broader system (Kahn and Roberds 2009).

The concept of the Exposure Amplification Factor (EAF), introduced in this thesis as an

original methodological contribution, captures the ratio of secondary systemic losses to direct default exposure — a metric that quantifies the extent to which settlement architecture transforms idiosyncratic institution failures into systemic events. To the author’s knowledge, no equivalent metric appears in the existing payment systems simulation literature, and its derivation forms part of the methodological contribution identified in Section 1.3.

Recent literature on Distributed Ledger Technology suggests that algorithmic consensus mechanisms may fundamentally alter this dynamic by providing near-instantaneous and immutable settlement. By reducing the time between transaction initiation and finality to 3–5 seconds, blockchain-based systems have the potential to minimise recognition lag and limit the scope for contagion propagation. The Ripple Protocol Consensus Algorithm (Schwartz et al. 2014) achieves deterministic finality through an iterative voting process among a Unique Node List of trusted validators. The simulation represents the residual operational risk that finality is not achieved within the settlement window through a stylised rollback probability $p_{\text{rollback}} = 10^{-4}$ per transaction — a deliberately small but non-zero baseline, consistent with the deterministic finality described by Schwartz et al. (2014) rather than an empirically reported figure. This near-zero rollback probability, combined with the 3–5 second finality window, effectively reduces the SF1–SF3 gap to zero by operational standards, eliminating the recognition lag that Berndsen (2018) identifies as the primary source of settlement contagion.

However, empirical evidence on the systemic stability implications of blockchain-based settlement remains limited. Shabsigh et al. (2020) survey central bank DLT experiments including Projects Jasper, Ubin, and Inthanon, and explicitly note that most experiments did not contain cost-benefit analyses or formal assessments of systemic resilience under stress conditions. This absence of empirical evidence motivates the stress testing analysis of Section 4.5, which compares Exposure Amplification Factors across five settlement mechanisms — RTGS, DNS, XRPL, wCBDC, and a hybrid RTGS–LSM design — under hub-default conditions.

Against these potential benefits, a balanced assessment requires acknowledging the distinct risk profile that blockchain-based settlement introduces. First, validator concentration risk: the XRPL Unique Node List creates a structural dependency on a small set of trusted validators, and coordinated validator failure or compromise would undermine the deterministic finality

guarantees that constitute the system's primary advantage (Chase and MacBrough 2018; Armknecht et al. 2015). This risk is structurally analogous to hub concentration in interbank payment networks — the same mechanism modelled in Section 4.5 — but operating at the consensus layer rather than the payment network layer. Second, governance risk: unlike TARGET2, which operates under ECB institutional governance with clear legal accountability, XRPL protocol parameters — including consensus thresholds, transaction fees, and the UNL composition itself — are subject to decentralised governance processes without a single accountable institution. The BIS Principles for Financial Market Infrastructures explicitly note that governance implemented solely through software is likely to be inflexible in a changing environment, since it is not feasible to consider all eventualities in software-based code established *ex ante* (Bank for International Settlements 2012). Shin (2026) identifies this as a fundamental fragmentation risk: decentralised coordination imposes structural costs on validators that centralised FMIs, operating under clear institutional mandates, do not face. Third, liquidity corridor risk: on-demand liquidity depends on market maker activity that may withdraw during stress conditions, precisely when settlement liquidity is most needed (Hernandez Cruz et al. 2024). Fourth, operational and cyber risk: while XRPL's architecture is simpler than smart-contract platforms, the BIS Principles for Financial Market Infrastructures include dedicated guidance on cyber resilience, requiring FMIs to undertake specific preparations and measures to limit the escalating risks that cyber threats pose to financial stability (Bank for International Settlements 2012). Furthermore, the PFMI application guidance explicitly notes that validating nodes operated by independent entities may have a bearing on operational risk, and that whether validating node operators constitute critical service providers depends on the design of the system and its rules, procedures and contractual arrangements (Bank for International Settlements 2012) — standards that XRPL has not yet formally demonstrated compliance with at institutional scale. These risks do not negate the efficiency and contagion advantages identified in this thesis, but they establish that the adoption of blockchain-based settlement involves a reallocation of risk — from the liquidity and latency risks of traditional architectures to the consensus, governance, and operational risks specific to distributed ledger systems — rather than a wholesale elimination

of systemic risk.

A distinct institutional alternative to permissionless blockchain settlement is wholesale central bank digital currency (wCBDC), which delivers the operational atomicity of distributed ledger settlement under central bank governance and within existing legal mandates. Wholesale CBDC pilots — most notably Project Helvetia III (Swiss National Bank 2023; Jordan 2024), with the ongoing Bank for International Settlements (2024) extending the approach to multi-currency cross-border settlement — demonstrate that the contagion-containment property of atomic settlement does not strictly require the validator-concentration and governance risks specific to permissionless blockchains: the same finality property can be delivered by a permissioned network whose validators are identified, regulated entities and whose settlement asset is central bank money. Alongside the conventional mechanisms (RTGS, DNS, and the hybrid RTGS–LSM design) and XRPL, the simulation framework therefore includes a permissioned wholesale CBDC as a dedicated analytical control. Because it settles atomically in central bank money, wCBDC shares XRPL’s operational finality properties — low latency, near-zero recognition lag, and contagion containment — while carrying none of the frictions specific to a permissionless ledger — slippage, XRP price and volatility exposure, and validator/governance risk. Comparing the two therefore measures which portion of XRPL’s performance is attributable to atomic settlement and which to its blockchain implementation. The XRPL calibration remains the empirical focus because price, volatility, and slippage data are publicly observable for XRPL and confidential for wCBDC pilots; wCBDC enters as a transparent benchmark rather than a separately calibrated system.

3.4 Cross-Border Settlement Frictions: Herstatt Risk and Correspondent Banking

Cross-border payment systems introduce additional layers of complexity, primarily due to differences in time zones, legal frameworks, and institutional arrangements. The most well-documented risk in this context is Herstatt risk, which arises from the temporal mismatch between payment legs in foreign exchange transactions (Berndsen 2018; Kahn and Roberds 2009). Named after the collapse of Bankhaus Herstatt in June 1974 — when German

regulators closed the bank after it had received Deutsche Mark payments from counterparties but before it had delivered the corresponding US dollar payments — Herstatt risk represents the principal risk that one party to a foreign exchange transaction delivers its currency leg without receiving the corresponding leg from its counterparty (Kahn and Roberds 2009).

To address this issue, the Continuous Linked Settlement (CLS) system was established in 2002, enabling payment-versus-payment (PvP) mechanisms that synchronise settlement across currencies (Berndsen 2018; Galati 2002). CLS operates through a 5-hour settlement window during which both legs of a foreign exchange transaction are settled simultaneously, reducing the Herstatt risk exposure window from 24 hours to 5 hours. While effective in reducing principal risk, CLS relies on limited operating windows and requires significant coordination between central banks, excluding numerous currency corridors from its coverage. As the simulation results of Section 5.4 demonstrate, CLS reduces Herstatt risk primarily through frequency reduction rather than severity mitigation: when defaults occur within the CLS window, conditional losses remain equal to the full transaction principal.

In parallel, the correspondent banking model continues to dominate cross-border payments. This system relies on pre-funded nostro and vostro accounts maintained by correspondent banks in foreign jurisdictions, leading to substantial amounts of idle liquidity distributed across jurisdictions. Bank for International Settlements (2020) estimates that hundreds of billions of euros are tied up in such arrangements on a daily basis, reflecting significant opportunity costs for financial institutions globally. The concentration of correspondent banking relationships has increased over the past decade, with the number of active correspondent banking relationships declining by approximately 25% between 2011 and 2020 (Committee on Payments and Market Infrastructures 2021), raising concerns about the resilience and inclusivity of the global payments architecture.

Emerging literature proposes the use of bridge assets and on-demand liquidity mechanisms as a potential alternative to pre-funded correspondent banking (Bank for International Settlements 2023b). By enabling just-in-time liquidity provisioning and continuous settlement, blockchain-based systems may reduce reliance on pre-funded accounts and improve capital efficiency. Bank for International Settlements (2023b) identifies three functional properties required

of a future monetary system: singleness of money, finality of settlement, and elasticity of supply. Blockchain-based systems such as the XRP Ledger address the first two properties through atomic settlement and deterministic finality, while the on-demand liquidity mechanism addresses the third through decentralised market-based liquidity provision. This thesis contributes to this debate by providing the first formal simulation-based assessment of these mechanisms within a comparative analytical framework.

3.5 Research Gap

The literature reviewed above establishes a robust understanding of the efficiency and risk properties of traditional settlement architectures, and a growing conceptual discussion of blockchain-based alternatives. However, three specific gaps remain unaddressed at the intersection of these two bodies of work.

First, agent-based simulation studies of RTGS systems, most notably Galbiati and Soramäki (2011), who derive a liquidity demand curve linking liquidity costs to settlement delay in a multi-agent RTGS framework, are confined to single-architecture settings. Their model isolates RTGS dynamics without comparing across settlement regimes or introducing a blockchain alternative.

Second, simulation-based efficiency frontier studies, including Leinonen and Soramäki (2005), who quantify the latency–liquidity trade-off across RTGS, DNS, and hybrid systems using Finnish payment data, predate blockchain-based settlement by two decades and do not address the question of whether DLT can shift rather than merely relocate along the efficiency frontier.

Third, institutional surveys of DLT experiments in payment systems, including Shabsigh et al. (2020), explicitly note that “most experiments did not contain cost-benefit analyses”, identifying the absence of formal comparative cost evaluation as an open research problem. This absence is particularly acute given recent theoretical work by Shin (2026), who identifies a fundamental scalability–coordination trade-off in decentralised networks that raises questions about whether blockchain-based settlement can deliver efficiency gains at institutional scale — questions that cannot be answered without the kind of simulation-based cost comparison this

thesis provides.

This thesis addresses all three gaps simultaneously by developing a unified simulation framework that combines RTGS, DNS, a hybrid RTGS–LSM design, and atomic settlement on both the XRP Ledger and wCBDC within a single Total Cost of Settlement function, deriving parametric switching conditions under which blockchain-based settlement dominates on both efficiency and systemic resilience grounds. The contribution is therefore methodological as well as empirical: it applies established simulation methods to a comparison that has not previously been attempted in the academic literature, and provides the first simulation-based formal quantitative evidence, to the author’s knowledge, on the conditions under which the XRP Ledger can serve as a credible alternative to traditional systemically important payment system architectures.

4 Methodology

This study adopts a quantitative, simulation-based comparative research design, combining network modelling with empirical calibration to evaluate the economic efficiency and risk characteristics of blockchain-based settlement systems relative to traditional Financial Market Infrastructures. The analysis is implemented through a discrete-time agent-based simulation framework, where alternative settlement architectures are modelled under controlled and comparable conditions. The empirical results are interpreted through the institutional lens of the Warehouse of Payments framework, which conceptualises payment systems as inventory management structures governed by trade-offs between liquidity, time, and risk.

4.1 Simulation Framework: Discrete-Time Agent-Based Modelling

Network Structure

The simulation network follows a scale-free Barabási–Albert topology with $n = 50$ nodes, comprising 5 hub nodes and 45 spoke nodes, motivated by empirical analyses documenting power-law degree distributions in interbank payment networks (Soramäki et al. 2007; Boss et al. 2004). This represents a 1:20 stylised scaling of TARGET2’s approximately 1,000 direct

participants (European Central Bank 2024), a scale sufficient for the purposes of this study because the qualitative cost rankings derived from the Total Cost of Settlement function are determined by the structural properties of each settlement architecture — namely liquidity requirements, settlement latency, and contagion mechanics — rather than by absolute network size. In a scale-free network, the probability that a node has k connections is proportional to $k^{-\gamma}$, where γ is the power-law exponent controlling the degree of hub dominance. The network is generated using the preferential attachment algorithm of Barabási and Albert (1999), with parameters $n = 50$, $\gamma = 2.5$, and $m = 2$ new edges per node. Under preferential attachment, nodes with higher degree accumulate new connections at a disproportionately faster rate — the “rich get richer” mechanism — producing a small number of highly connected hubs and a large number of lightly connected spokes. The 5 hub nodes represent systemically important institutions that intermediate the majority of payment flows, while the 45 spoke nodes represent smaller participants that route most payments through the hubs.

The resulting network exhibits a hub-to-spoke degree ratio of 23.5, with the highest-degree hub node connected to 47 of the 49 remaining nodes. This concentration is consistent with the empirical finding of Soramäki et al. (2007) that interbank payment networks exhibit extreme connectivity concentration, with a small number of systemically important institutions handling the majority of payment flows. Specifically, Soramäki et al. (2007) document that the Fedwire interbank payment network contains a tightly connected core of money-centre banks to which most other banks connect, with a scale-free degree distribution over a substantial range — a structural property of hub concentration that this simulation replicates at stylised scale through the hub-to-spoke degree ratio, without claiming formal distributional equivalence at $n = 50$. Due to the stylised network size of 50 nodes, formal power-law fitting via maximum likelihood estimation is not statistically reliable; network concentration is therefore validated through the hub-to-spoke degree ratio rather than direct estimation of γ (Clauset et al. 2009). Sensitivity analysis over $\gamma \in \{2.0, 2.5, 3.0\}$, implemented by varying the `power` parameter of the Barabási–Albert algorithm, confirms that the qualitative simulation results are robust to this calibration choice, with the hub-to-spoke ratio remaining stable between 23.5 and 24.0 across all tested values.

Payment Arrival Process

Payment arrivals follow a Poisson process with rate parameter λ , reflecting the standard modelling assumption for large-value payment systems in the agent-based simulation literature (Galbiati and Soramäki 2011). The Poisson process models the arrival of independent payment events over time: at each discrete time step t , the total number of payments arriving across the network is drawn from a Poisson distribution with aggregate rate λ , with sender and receiver pairs then assigned by sampling nodes with probability proportional to degree. This generates random but statistically predictable payment flows, consistent with the empirical observation that large-value payment arrivals are approximately memoryless within intraday operating windows.

The simulation runs across $T = 32$ discrete time steps of $\Delta t = 0.25$ hours each (15-minute intervals), spanning the eight-hour steady-state core of the settlement day. An RTGS operating day is not temporally homogeneous: it comprises a start-of-day phase (liquidity provisioning and reserve positioning), a steady-state phase of continuous real-time gross settlement, and an end-of-day phase (queue resolution, liquidity-saving offsetting, and reserve squaring). Only the steady-state phase reflects the settlement mechanics this study compares; the boundary phases are institutional artefacts of the discrete RTGS operating day that have no analogue in an always-on ledger such as XRPL, and only a partial one in the remaining mechanisms. Modelling them would inject RTGS-specific dynamics into a comparison the other architectures cannot structurally exhibit. The simulation therefore abstracts from both the start-of-day provisioning phase and the end-of-day closing procedure; the residual steady-state window — approximately eight hours, consistent with the daytime processing window of a TARGET2 business day and with stylised agent-based frameworks in the payment systems literature (Galbiati and Soramäki 2011; Leinonen and Soramäki 2005) — is the period simulated. This discretisation remains computationally tractable while providing sufficient granularity to capture intraday payment dynamics across all settlement architectures. The baseline daily transaction volume is $\Lambda = 17,500$ payments per day, derived from TARGET2's historical average of approximately 350,000 payments per day (Deutsche Bundesbank 2023) scaled by the same 1:20 factor applied consistently across all network and liquidity parameters

in this framework ($350,000 \div 20 = 17,500$). While linear scaling of payment volume with participant count is a simplifying assumption — volumes in scale-free networks may scale non-linearly with node count due to hub concentration effects — the 1:20 factor is applied uniformly across network size, transaction volume, and liquidity requirements, preserving the internal consistency of the simulation framework and ensuring proportional comparability across all settlement architectures. The lower bound of 10,000 daily transactions is set conservatively below this proportional baseline, ensuring the network operates in a sparse regime consistent with the stylised nature of the simulation. The upper bound of 100,000 represents a high-stress volume scenario approximately six times the proportional baseline, designed to test whether the cost ordering holds under substantially higher transaction loads than the baseline calibration.

Payment amounts $P_{ij}(t)$ follow a log-normal distribution:

$$\ln P_{ij}(t) \sim \mathcal{N}(\mu_{\log}, \sigma_{\log}^2), \quad \mu_{\log} = 5, \quad \sigma_{\log} = 1.5$$

The theoretical mean and median are derived from standard log-normal moment formulae:

$$\mathbb{E}[P] = \exp\left(\mu_{\log} + \frac{\sigma_{\log}^2}{2}\right) = \exp\left(5 + \frac{1.5^2}{2}\right) = \exp(6.125) \approx \text{€}472$$

$$\text{Median}[P] = \exp(\mu_{\log}) = \exp(5) \approx \text{€}149$$

generating a right-skewed distribution consistent with the empirical observation that large-value payment systems process a small number of very large transactions alongside a large volume of smaller ones. Aggregating across the daily payment population, the expected total network value is $\lambda \times \mathbb{E}[P] \approx 17,500 \times \text{€}472 \approx \text{€}8.3$ million per day; the realised value at the seed used in this thesis is approximately $\text{€}7.47$ million, reflecting Poisson variation and the discarding of self-payments. This stylised aggregate is the anchor against which the 30% RTGS prefunding ratio (Section 4.2) and the 30.6% XRPL gross-pool calibration are defined. Payment senders and receivers are drawn from the network with probability proportional to node degree, ensuring that hub nodes naturally send and receive more payments than spoke

nodes, consistent with the empirical topology documented by Soramäki et al. (2007).

Formal Specification of the Simulation Loop

Let $\mathcal{N} = \{1, 2, \dots, n\}$ denote the set of $n = 50$ participating institutions, and let $t \in \{1, 2, \dots, T\}$ with $T = 32$ index discrete time steps of $\Delta t = 0.25$ hours each (15-minute intervals), spanning 8 operating hours. The system state at each step is characterised by the vector of liquidity balances:

$$\mathbf{L}(t) = (L_1(t), L_2(t), \dots, L_n(t)), \quad L_i(t) \geq 0 \quad \forall i \in \mathcal{N}$$

At each time step, payment instructions arrive according to:

$$N(t) \sim \text{Poisson}(\lambda), \quad \text{sender}_k, \text{receiver}_k \sim p_i \propto \deg(i), \quad k = 1, \dots, N(t)$$

where $N(t)$ is the total number of payments arriving at step t , and each payment is assigned a sender and receiver by independent draws proportional to node degree, with self-payments discarded.

4.2 Settlement Architecture Specifications

Five settlement configurations are implemented as counterfactual scenarios within the unified simulation framework: three core architectures (RTGS, DNS, and XRPL), a hybrid RTGS–LSM benchmark, and a permissioned wholesale CBDC control. All operate on the same payment population and network structure, ensuring that differences in results are attributable solely to settlement architecture rather than to different payment draws or network configurations. For each scenario, the formal settlement rule and liquidity update are presented alongside the institutional and regulatory context that motivates the specification.

Scenario 1: Real-Time Gross Settlement (RTGS)

The RTGS scenario replicates the operational structure of TARGET2, the Eurosystem’s large-value payment system (European Central Bank 2010). In TARGET2, transactions

are processed individually and settled in central bank money on a continuous gross basis throughout the operating day. Settlement is immediate and unconditional provided the sending institution holds sufficient intraday liquidity, which is typically obtained through collateralised intraday credit from the central bank.

Each payment $P_{ij}(t)$ is processed individually upon arrival. Settlement occurs if and only if the sending node holds sufficient liquidity:

$$\text{Settle}_{ij}^{\text{RTGS}}(t) = \begin{cases} 1 & \text{if } L_i(t) \geq P_{ij}(t) \\ 0 & \text{(queued, retried in FIFO order)} \end{cases}$$

Unsettled payments join a first-in-first-out queue and are retried at each subsequent time step until either settled or the operating day ends. Liquidity balances update at each step as:

$$L_i(t+1) = L_i(t) - \sum_{j \neq i} \mathbf{1}[\text{Settle}_{ij}^{\text{RTGS}}(t) = 1] P_{ij}(t) + \sum_{j \neq i} \mathbf{1}[\text{Settle}_{ji}^{\text{RTGS}}(t) = 1] P_{ji}(t)$$

where the first sum captures outflows from settled payments sent by institution i and the second captures inflows from settled payments received by institution i . The indicator function $\mathbf{1}[\cdot]$ equals 1 when the condition in brackets holds and 0 otherwise.

A partial liquidity recycling delay parameter of 0.421 is incorporated, representing the fraction of received funds not immediately available for outgoing payments within the same time step. This reflects the empirical friction that collateral mobilisation lags and intraday credit processing times introduce in TARGET2 (European Central Bank 2010). The parameter was calibrated under a coarser 8-step discretisation (one hour per step) by minimising the squared deviation between the simulated gridlock probability and the TARGET2 benchmark of approximately 15% under 3x stress volume, producing a simulated gridlock probability of 14.80%. It is carried forward to the 32-step framework because re-calibration at this granularity is infeasible: per-step payment volume is too thin for gridlock to manifest as a simulation observable. The limitation is one of transferability across discretisations rather than arbitrary parameter choice — the value is empirically grounded but its validity at finer

granularity cannot be confirmed from simulation evidence alone. See Section 6.1 for a discussion of the direction of bias.

The required safety stock for each institution incorporates the 2% collateral haircut on intraday credit applied by the Eurosystem to eligible collateral (European Central Bank 2010; Berndsen 2018):

$$L_i^{\text{RTGS}} = \frac{\text{Peak outflow}_i}{1 - 0.02}$$

Total prefunded liquidity is set at 30% of total daily payment value, calibrated to the TARGET2 operational structure in which participating institutions maintain intraday credit lines equivalent to approximately one third of their daily payment obligations (European Central Bank 2010). The aggregate liquidity requirement is $L^{\text{RTGS}} = \sum_{i \in \mathcal{N}} L_i^{\text{RTGS}}$, and settlement latency D^{RTGS} is the average queuing time across all transactions, computed empirically from the simulation as 0.0026 hours.

Scenario 2: Deferred Net Settlement (DNS)

The DNS scenario models the netting-based architecture of systems such as EURO1, the private large-value payment system operated by the Euro Banking Association (Berndsen 2018). In DNS systems, payment instructions accumulate throughout a clearing cycle and only multilateral net positions are settled at the end of each cycle. This recovers liquidity efficiency relative to RTGS — because offsetting payment flows cancel out and only the residual net obligation needs to be funded — but reintroduces settlement delays and counterparty exposure during the SF1–SF3 settlement gap (Berndsen 2018).

Payments accumulate during clearing cycles of fixed duration $\tau = 2$ hours (equal to 8 time steps), consistent with the EURO1 operating schedule (Manning et al. 2009). No settlement occurs within the cycle. At the end of each cycle, multilateral net positions are computed for every institution:

$$\text{Net}_i(\tau) = \sum_{j \neq i} P_{ji}(\tau) - \sum_{j \neq i} P_{ij}(\tau)$$

where the first term aggregates all inflows received by institution i during the cycle and the second term aggregates all outflows sent by institution i . Settlement is executed only for

net payers ($\text{Net}_i(\tau) < 0$), who must fund their net obligation at cycle end. Net receivers ($\text{Net}_i(\tau) > 0$) receive their net inflow without prefunding.

During the SF1–SF3 settlement gap, net receivables held by participants against net debtors constitute unsecured short-term interbank credit exposures in the sense of the Basel II standardised approach (BCBS 2006; Manning et al. 2009): bank Y , having credited its customer, holds a net receivable from bank X that is unsecured and subject to counterparty default risk until end-of-cycle settlement (Berndsen 2018). The required safety stock is therefore derived from the Basel II standardised approach risk weight of 20% applied to short-term interbank exposures, multiplied by the 8% minimum capital adequacy ratio (BCBS 2006), yielding a credit risk premium of $0.20 \times 0.08 = 1.6\%$ on residual net exposures:

$$L^{\text{DNS}} = \frac{\sum_{i \in \mathcal{N}} \max(0, -\text{Net}_i(\tau))}{1 - 0.016}$$

The denominator $1 - 0.016 = 0.984$ reflects the fact that institutions must hold sufficient safety stock to cover their net obligations after accounting for the 1.6% credit risk premium: to fund a net obligation of $\text{€}X$, an institution must prefund $\text{€}X/0.984$. This parameter is distinct from the 2% collateral haircut applied in the RTGS scenario, which reflects a valuation margin on central bank eligible collateral rather than a credit risk premium (European Central Bank 2010). Using the regulatory minimum of 1.6% rather than a higher empirical estimate represents a conservative calibration that favours DNS performance relative to XRPL — a higher credit risk premium would increase L^{DNS} and therefore DNS total settlement costs. The 1.6% figure therefore constitutes the most demanding test of the cost ordering under the Basel II standardised approach (BCBS 2006).

The endogenous liquidity saving ratio is:

$$\text{LSR} = 1 - \frac{L^{\text{DNS}}}{L^{\text{RTGS}}}$$

validated against the BoF-PSS2 benchmark of 82% (Leinonen and Soramäki 2003), with the simulation producing an LSR of 82.66%, confirming calibration validity. Settlement latency

$D^{\text{DNS}} = 0.8881$ hours on average, reflecting the expected residual waiting time within the 2-hour clearing cycle. Under a continuous arrival process, the theoretical expected wait would be $\tau/2 = 1$ hour (Manning et al. 2009); the 32-step discretisation produces 0.8881 hours, representing 88.8% of the theoretical benchmark.

Scenario 3: Hybrid RTGS with Liquidity Saving Mechanism

The hybrid scenario combines the continuous finality of RTGS with the multilateral offsetting of DNS, the liquidity saving mechanism deployed in modern systems such as TARGET2 (Manning et al. 2009). Queued payments are multilaterally netted at fixed offsetting intervals of one operating hour (four time steps) — half the DNS clearing cycle — and only the residual net positions are funded, recovering part of the DNS liquidity saving while preserving near-RTGS finality. The liquidity requirement is the net funding used across offsetting rounds, $L^{\text{hybrid}} = \sum_{\text{rounds}} \max(0, -\text{Net}) / (1 - p)$, where $p = 0.016$ is the Basel II credit-risk premium applied to short-term interbank exposures within each offsetting round, identical to the DNS premium of Section 4.2, since queued payments remain exposed to counterparty credit risk during the one-hour accumulation window before each netting cycle, albeit over a shorter horizon than DNS, and settlement latency is the average residual wait within the one-hour offsetting window. By construction the hybrid occupies the interior of the latency–liquidity plane between the pure RTGS and pure DNS endpoints, providing the strongest traditional benchmark against which the atomic architectures are assessed. The simulation produces an average settlement latency of $D^{\text{hybrid}} = 0.3781$ hours — intermediate between RTGS (0.0026 hours) and DNS (0.8881 hours) — confirming the hybrid’s interior position on the latency–liquidity plane. The corresponding liquidity requirement and total cost are reported in Table 2.

Scenario 4: Blockchain-Based Atomic Settlement (XRPL)

The XRPL scenario models the settlement architecture of the XRP Ledger, a public blockchain network designed for fast and low-cost cross-border payments through atomic settlement and on-demand liquidity provisioning (XRP Ledger Foundation 2026). Unlike RTGS and DNS,

which rely on pre-funded central bank or commercial bank balances, XRPL settles payments atomically in 3–5 seconds through the Ripple Protocol Consensus Algorithm, achieving deterministic finality through an iterative voting process among a Unique Node List of trusted validators (Schwartz et al. 2014).

Liquidity is provided through an on-demand mechanism rather than pre-funded balances. In Ripple’s On-Demand Liquidity product, cross-border payments are executed by purchasing XRP on a source exchange, transferring it across the XRP Ledger, and selling it on a destination exchange. The effective liquidity available for a given payment therefore depends on the combined order book depth of two separate exchanges at opposite ends of the payment corridor, which varies by corridor, exchange, time of day, payment size, and prevailing market maker activity. Unlike AMM-based decentralised exchanges, the XRP Ledger does not rely on a unified liquidity pool: liquidity is fragmented across multiple order books and aggregated dynamically through a pathfinding algorithm (XRP Ledger Foundation 2026). As a result, market depth is not directly observable as a single metric and must be modelled as a liquidity parameter within the simulation framework.

Each payment $P_{ij}(t)$ settles atomically within 3–5 seconds or fails entirely, with no queuing:

$$\text{Settle}_{ij}^{\text{XRPL}}(t) = \begin{cases} 1 & \text{if } \text{MD}(t) \geq P_{ij}(t) \\ 0 & \text{(failed, no retry)} \end{cases}$$

where $\text{MD}(t)$ is the available pool depth at time t . Unlike RTGS where failed payments queue for retry, XRPL failures are final: each payment either achieves atomic finality within the 3–5 second window or is rolled back entirely, with a stylised rollback probability $p_{\text{rollback}} = 10^{-4}$ per transaction (a small but non-zero baseline adopted in this study; see Section 3.3). This near-zero rollback probability effectively reduces the SF1–SF3 settlement gap to zero by operational standards, eliminating the recognition lag that Berndsen (2018) identifies as the primary source of settlement contagion.

The slippage cost on each settled transaction captures the market impact of routing a payment

through the on-demand liquidity pool:

$$\text{Slippage}_{ij}(t) = \kappa \cdot \left(\frac{P_{ij}(t)}{\text{MD}(t)} \right)^\delta \cdot P_{ij}(t), \quad \kappa = 0.05, \quad \delta = 1.5$$

The slippage parameters $\kappa = 0.05$ and $\delta = 1.5$ are calibrated to be consistent with the empirical slippage distribution documented by Hernandez Cruz et al. (2024), who report an 80th percentile slippage range of 1.36%–1.73% on the XRPL decentralised exchange. Neither parameter is directly reported as a point estimate in that paper. $\kappa = 0.05$ is reverse-engineered as the scaling coefficient that reproduces outputs within the observed slippage range at the Hernandez Cruz observed trade size of €955,675 against the institutional-scale gross pool value of $(MD \times P_{\text{XRP}}) \times 20 = €2,287,349 \times 20 = €45,746,980$, applying the 1:20 scaling consistently across this framework. Specifically:

$$\kappa = \frac{\text{Slippage}\%}{\left(\frac{P}{\text{MD}_{\text{institutional}}} \right)^\delta} = \frac{0.01545}{\left(\frac{955,675}{45,746,980} \right)^{1.5}} = \frac{0.01545}{0.003019} \approx 0.051 \approx 0.05$$

where 0.01545 is the midpoint of the observed 80th percentile slippage range of 1.36%–1.73%. $\delta = 1.5$ is a theoretically motivated intermediate value: it is above unity, capturing the empirically documented convexity of price impact with respect to transaction size relative to pool depth, while remaining sub-quadratic, consistent with the moderate rather than extreme market impact documented in Hernandez Cruz et al. (2024) for the XRPL DEX under normal operating conditions. Sensitivity analysis over $\kappa \in \{0.03, 0.04, 0.05, 0.06, 0.07\}$ and $\delta \in \{1.0, 1.5, 2.0\}$ confirms that the qualitative cost ordering is robust across the full range of plausible parameter values.

Consistent with the symmetric capital treatment of Section 3.2, two liquidity concepts are defined for XRPL. Under the primary gross-capital specification, the liquidity requirement is the full gross pool value:

$$L_{\text{gross}}^{\text{XRPL}} = MD \times P_{\text{XRP}} = 1,971,853 \times 1.16 = €2,287,349$$

charged the full opportunity cost exactly as RTGS prefunding is. Under the risk-adjusted

alternative, the requirement is reduced to the volatility buffer — the capital at risk from XRP price movements during the 3–5 second settlement window:

$$L_{\text{buffer}}^{\text{XRPL}} = MD \times \sigma_{\text{XRP}} \times P_{\text{XRP}} = 1,971,853 \times 0.05 \times 1.16 = \text{€}114,367$$

where $MD = 1,971,853$ XRP is the effective market depth parameter calibrated so that the gross pool value represents approximately 30.6% of total daily network payment value — consistent with the 30% prefunding ratio applied in the RTGS scenario. Using the volatility buffer rather than the gross pool as the risk-adjusted liquidity input requires theoretical justification. In RTGS, prefunded balances represent capital genuinely locked as central bank collateral throughout the operating day — it cannot be deployed elsewhere and its full value carries an opportunity cost specific to the settlement function. The gross XRPL pool, by contrast, resides in exchange order books and is not institutionally encumbered in the sense of RTGS collateral: it remains tradeable and generates market making returns. While the full pool value does carry an opportunity cost in the broad sense — capital deployed to an ODL corridor cannot simultaneously be deployed elsewhere — this cost is offset by market making fee income and is not specific to the settlement function. The capital cost *attributable to settlement* is therefore limited to the volatility buffer $MD \times \sigma_{\text{XRP}} \times P_{\text{XRP}}$: the expected loss from adverse XRP price movements during the 3–5 second settlement window, which cannot be offset by any market making activity occurring on a longer timescale. This distinction mirrors the role of initial margin in derivatives markets, which covers potential adverse price moves over the close-out horizon rather than the full notional value of the position — the same logic applied here to a 3–5 second settlement window. The gross pool value $MD \times P_{\text{XRP}} = \text{€}2,287,349$ serves a distinct role, reported in Table 7 as the gross-corridor comparator against nostro prefunding rather than as a risk-adjusted capital cost.

$\sigma_{\text{XRP}} = 5\%$ is the daily XRP price volatility, adopted as a conservative estimate above the two-year realised average of 4.46% (CoinGecko 2026), and $P_{\text{XRP}} = \text{€}1.16$ is the XRP/EUR exchange rate verified against CoinGecko data as of March 2026 (CoinGecko 2026). Settlement latency is $D^{\text{XRPL}} \approx 0$ by construction, with average finality of approximately 4 seconds

drawn uniformly from the 3–5 second window.

Scenario 5: Permissioned Wholesale CBDC (wCBDC)

The wCBDC scenario models atomic payment-versus-payment settlement in central bank money on a permissioned distributed ledger operated by central banks. It shares XRPL’s operational properties — atomic finality within seconds, no queuing, near-zero recognition lag — but differs in its liquidity economics. Settlement occurs in the domestic settlement currency, so there is no FX volatility exposure and hence no volatility buffer; and transfers are atomic on a permissioned ledger rather than routed through external order books, so there is no slippage. The corridor is endowed, by construction, with central bank money equal in value to the XRPL gross pool, so that any divergence in total cost between wCBDC and XRPL is attributable solely to the public-blockchain features rather than to atomic settlement, which both share:

$$L^{\text{wCBDC}} := L_{\text{gross}}^{\text{XRPL}} = \text{€}2,287,349, \quad \Sigma_{\text{slip}}^{\text{wCBDC}} = 0, \quad D^{\text{wCBDC}} \approx 0.$$

The definitional equality $:=$ signals that wCBDC’s liquidity requirement is a control-design choice (matching the XRPL gross pool to enable like-for-like comparison) rather than a quantity that inherits exposure to the XRP exchange rate; wCBDC is denominated in central bank money throughout and is invariant to P_{XRP} .

4.3 Total Cost of Settlement Function

The unified Total Cost of Settlement (TCS) function enables direct comparison across all architectures:

$$\text{TCS} = (L \times r) + \varphi(D), \quad \varphi(D) = \alpha D^\beta \quad (1)$$

where L is the total liquidity requirement, r is the opportunity cost of capital, D is the average settlement latency, $\alpha = 0.1$ is the delay cost scaling parameter, and $\beta = 1.8$ is the delay cost

convexity parameter (Bech and Garratt 2003). Sensitivity analysis over $\alpha \in \{0.05, 0.1, 0.2\}$ and $\beta \in \{1.2, 1.8, 2.5\}$ confirms robustness.

For the XRPL scenario, an additional slippage cost term Σ_{slip} is included to capture the market impact of routing payments through the on-demand liquidity pool:

$$\text{TCS}^{\text{XRPL}} = (L^{\text{XRPL}} \times r) + \varphi(D^{\text{XRPL}}) + \Sigma_{\text{slip}}, \quad \Sigma_{\text{slip}} = \sum_i \kappa \cdot \left(\frac{P_i}{\text{MD}} \right)^\delta \cdot P_i \quad (2)$$

where L^{XRPL} denotes the liquidity input under whichever concept is applied: $L_{\text{gross}}^{\text{XRPL}}$ under the primary gross-capital specification, or $L_{\text{buffer}}^{\text{XRPL}}$ under the risk-adjusted alternative (Section 4.2). At baseline parameters ($\kappa = 0.05$, $\delta = 1.5$, $MD = 1,971,853$ XRP, $\lambda = 17,500$), the aggregate slippage cost evaluates to $\Sigma_{\text{slip}} = \text{€}80$. Its weight in total cost is therefore specification-dependent: it is negligible (0.12%) under the gross-capital opportunity cost of $L_{\text{gross}}^{\text{XRPL}} \times r = \text{€}68,620$, and a small but larger share (2.28%) under the risk-adjusted buffer cost of $L_{\text{buffer}}^{\text{XRPL}} \times r = \text{€}3,431$. Under either concept the liquidity opportunity cost, not slippage, is the dominant cost driver. RTGS and DNS do not incur slippage costs, as settlement occurs through prefunded central bank or commercial bank balances rather than market-based liquidity pools.

4.4 Efficiency Frontier Analysis

The efficiency frontier is constructed in the latency–liquidity plane (D, L) . XRPL shifts the frontier if and only if:

$$D_{\text{XRPL}} < \min(D_{\text{RTGS}}, D_{\text{DNS}}) \quad \text{and} \quad L_{\text{XRPL}} < \min(L_{\text{RTGS}}, L_{\text{DNS}}) \quad (3)$$

The threshold interest rate r^* and transaction volume λ^* are identified through parametric sensitivity analysis as the conditions under which XRPL's TCS falls below that of both RTGS and DNS.

4.5 Stress Testing and Contagion Analysis

Stress testing introduces an exogenous default of the highest-degree hub node (node 1, degree = 47) at peak volume ($\lambda = 50,000$, approximately three times the baseline daily volume) and at the midpoint of the operating day (step $t = 16$ corresponding to 4 operating hours), representing the moment of maximum systemic pressure when intraday liquidity recycling is most active. The stress volume of 50,000 is distinct from the upper bound of 100,000 used in the sensitivity analysis of Section 4.4, which tests XRPL volume dominance conditions rather than contagion dynamics. The Exposure Amplification Factor is:

$$\text{EAF} = \frac{\text{Secondary systemic exposure}}{\text{Direct hub exposure}} \quad (4)$$

Affected nodes are defined as all counterparty institutions that sustain any loss of expected inflows as a result of the hub default, regardless of whether those losses are sufficient to cause insolvency. This definition is consistent with the EAF formula, which measures total secondary exposure rather than the subset of exposures that exceed institutional liquidity buffers. Under this definition, the near-complete connectivity of the defaulting hub (degree 47 out of 49 possible connections) implies that virtually all network participants are exposed to some degree of secondary loss, producing the high affected node counts observed in the RTGS and XRPL scenarios. The DNS scenario produces fewer affected nodes (24) because netting partially offsets bilateral exposures within the clearing cycle — nodes that both sent and received payments to and from the hub have their gross exposure reduced to a net receivable, and only those with a positive net receivable are counted as affected. Monte Carlo robustness is ensured via 1,000 independent runs, with all standard errors confirmed below 5%.

4.6 Cross-Border Extension

Herstatt risk is modelled as the expected loss on a single large-value cross-border payment of $P = \text{€}1,000,000$, representing a stylised wholesale transaction consistent with the large-value nature of the correspondent banking and FMI settlement systems under comparison (Bank for

International Settlements 2020), at a scale where principal risk — defined as exposure to the full transaction value in the event of counterparty default (Berndsen 2018) — is economically significant. The expected loss formula captures the probability that a counterparty default occurs during the settlement gap Δt , following the standard expected-loss decomposition of exposure and default probability (BCBS 2006), with default modelled as a constant-hazard process over the gap:

$$\mathbb{E}[\text{Loss}] = P \cdot (1 - (1 - p)^{\Delta t}) \quad (5)$$

where $p = 10^{-4}/\text{hour}$ is the stylised baseline default probability per hour and $\Delta t \in \{24, 5, 0\}$ hours for correspondent banking, CLS, and XRPL respectively. The three values of Δt correspond to the full 24-hour time zone exposure under correspondent banking, the 5-hour CLS settlement window, and the effectively zero settlement gap under XRPL atomic settlement. The default probability $p = 10^{-4}/\text{hour}$ is a stylised baseline calibrated to produce default rates in the range empirically documented for large-value payment system participants, and is held constant across all three architectures so that differences in expected loss are attributable solely to the settlement gap Δt rather than to differences in counterparty credit quality.

The pool depth threshold MD^* is derived as the point at which XRPL on-demand liquidity cost equals the opportunity cost of maintaining correspondent nostro balances:

$$MD^* = \frac{\text{Nostro prefunding}}{P_{XRPL}} = \frac{\text{€}200,000,000}{\text{€}1.16} = 172,413,793 \text{ XRP} \quad (6)$$

where €200,000,000 represents the stylised nostro prefunding requirement for a representative cross-border payment corridor, consistent with the BIS estimate that hundreds of billions of euros are held in nostro and vostro accounts globally (Bank for International Settlements 2020).

Cross-border results are validated through 100,000 Monte Carlo runs rather than the 1,000 used elsewhere. This larger sample is required because Herstatt default events are rare by construction — occurring with probability 0.24% and 0.06% under correspondent banking and CLS respectively — meaning that with only 1,000 runs the expected number of observed

default events would be 2.4 and 0.6 respectively, which is insufficient for stable default rate estimation. With 100,000 runs the expected default counts rise to 240 and 60, providing adequate observations to confirm all default rate standard errors below 0.05%. This approach is consistent with standard simulation practice for rare event estimation, where the required sample size scales with the inverse of the event probability rather than with the complexity of the model.

4.7 Data Sources and Calibration

Table 1 summarises all simulation parameters, their baseline values, and calibration sources.

Table 1: Simulation Parameters and Calibration Sources

Parameter	Value	Description	Source
n	50	Network nodes	Stylised
γ	2.5	Power-law exponent	Soramäki et al. (2007)
λ	17,500	Baseline daily payments	Deutsche Bundesbank (2023)
T	32	Discrete time steps	This study
Δt	0.25 hrs	Step length (15-minute intervals)	This study
r	3%	Baseline interest rate	Baseline within tested range [0%,8%]
α	0.1	Delay cost parameter	Bech and Garratt (2003)
β	1.8	Delay cost convexity	Bech and Garratt (2003)
Haircut	2%	Collateral haircut	European Central Bank (2010)
recycle_delay	0.421	RTGS within-step liquidity friction	Calibrated under 8-step discretisation; see Section 6.1
Credit premium	1.6%	DNS & Hybrid credit risk premium	BCBS (2006)
LSR target	82%	DNS liquidity saving	Leinonen and Soramäki (2003)
p_{rollback}	10^{-4}	XRPL rollback probability	Stylised baseline
Finality	3–5s	XRPL settlement latency	XRP Ledger Foundation (2026)
κ	0.05	Slippage coefficient	Reverse-engineered from Hernandez Cruz et al. (2024)
δ	1.5	Slippage convexity	Hernandez Cruz et al. (2024)
σ_{XRP}	5%	XRP daily volatility	Author's calculation, CoinGecko (2026)
P_{XRP}	€1.16	XRP/EUR exchange rate	CoinGecko (2026)
p	$10^{-4}/\text{hr}$	Default probability	Stylised baseline
MC runs	1,000/100,000	Replications	Section 4.1

5 Results

5.1 Baseline Settlement Costs

Table 2 presents the baseline simulation results for all five settlement architectures under identical network conditions: a 50-node scale-free payment network, daily transaction volume of $\lambda = 17,500$ payments, and interest rate $r = 3\%$. XRPL is reported under two liquidity concepts to make the framing dependence of the headline result transparent at baseline rather than deferring it to the efficiency frontier of Section 5.2.

Table 2: Baseline Settlement Cost Comparison under the gross-capital specification ($\lambda = 17,500$, $r = 3\%$, $P_{XRP} = \text{€}1.16$). The volatility-buffer column reports the risk-adjusted alternative for XRPL only.

Metric	RTGS	DNS	Hybrid	wCBDC	XRPL	
					Gross	Vol. Buffer
Liquidity requirement (L)	€2,287,265	€1,315,335	€1,712,511	€2,287,349	€2,287,349	€114,367
Settlement latency (D , hrs)	0.0026	0.8881	0.3781	0.0011	0.0011	0.0011
Settlement rate	99.80%	100%	100%	100%	99.99%	99.99%
Liquidity Saving Ratio vs RTGS	—	42.5%	25.1%	—	—	—
Opportunity cost ($L \times r$)	€68,618	€39,460	€51,375	€68,620	€68,620	€3,431
Delay penalty $\varphi(D)$	€0.00	€0.08	€0.02	≈ 0	≈ 0	≈ 0
Slippage Σ_{slip}	—	—	—	—	€80	€80
Total Cost of Settlement	€68,618	€39,460	€51,375	€68,620	€68,701	€3,511

Notes: Under the primary gross-capital specification every architecture is charged the full opportunity cost of deployed capital. The volatility-buffer column costs only XRPL’s settlement-window capital at risk and is the risk-adjusted alternative. wCBDC settles atomically in central bank money (gross pool, no slippage, no buffer), so its TCS coincides with RTGS and isolates the atomic-settlement effect from the buffer accounting. The hybrid nets queued payments every operating hour, occupying the interior between RTGS and DNS.

The two XRPL columns report the same simulation under alternative liquidity concepts. The volatility-buffer concept treats $L_{XRPL} = MD \times \sigma_{XRP} \times P_{XRP} = \text{€}114,367$ as the capital at risk from XRP price movements during the 3–5 second settlement window — the settlement-attributable opportunity cost that enters the TCS function. The gross-pool concept reports $MD \times P_{XRP} = \text{€}2,287,349$, the gross capital deployed to the corridor, which is the appropriate comparator against RTGS prefunding and correspondent nostro balances as a capital-deployment measure. The gross pool is not institutionally encumbered in the sense of RTGS collateral and generates market making returns, which is the theoretical basis for excluding it from the TCS function (Section 4.2).

Under the volatility-buffer concept, the TCS comparison yields a clear cost ordering:

$$\text{TCS}_{\text{XRPL}} < \text{TCS}_{\text{DNS}} < \text{TCS}_{\text{RTGS}} \Rightarrow \text{€}3,511 < \text{€}39,460 < \text{€}68,618. \quad (7)$$

Under the gross-pool comparator, XRPL matches RTGS on liquidity ($\text{€}2,287,349 \approx \text{€}2,287,265$) but retains the latency advantage of 0.0011 hours against 0.0026 hours. The latency dimension of the frontier shift identified in Section 5.2 is therefore unconditional on the liquidity concept; the liquidity dimension is conditional on the volatility buffer being accepted as the settlement-attributable cost. The remainder of this section adopts the gross-capital ordering

as the primary result and reports the volatility-buffer ordering alongside as the risk-adjusted alternative; the distinction is made explicit at every subsequent result.

Finding 1: Liquidity Costs dominate Total Settlement Costs

Across all five configurations, the opportunity cost of prefunded liquidity ($L \times r$) accounts for virtually the entire Total Cost of Settlement, with delay penalties $\varphi(D) = \alpha D^\beta$ representing negligible second-order frictions. Under RTGS, the delay penalty evaluates to $\varphi(D^{\text{RTGS}}) = 0.1 \times (0.0026)^{1.8} \approx \text{€}9.3 \times 10^{-7}$, rounding to €0.00 at any practical precision — a consequence of raising a queuing time of order 10^{-3} hours to the power 1.8 under 15-minute discretisation, discussed further in Section 6.1. Under DNS, the delay penalty of €0.08 represents 0.0002% of total TCS — despite a settlement latency of 0.8881 hours, the convex delay cost function $\varphi(D) = 0.1 \times (0.8881)^{1.8} \approx \text{€}0.08$ produces a negligible absolute cost because $\alpha = 0.1$ is calibrated to ensure delay penalties remain second-order relative to liquidity costs at baseline interest rates, consistent with the parameter motivation in Section 4.3. The hybrid configuration, with an intermediate latency of 0.3781 hours, yields $\varphi(D^{\text{hybrid}}) \approx \text{€}0.02$ — still under 0.00004% of its total settlement cost — confirming that the principle holds across the full latency range modelled. Under XRPL, the delay penalty is effectively zero ($\varphi(D) \approx 4.8 \times 10^{-7}$), reflecting 3–5 second atomic finality (XRP Ledger Foundation 2026); the wCBDC scenario inherits the same near-zero penalty by construction, sharing atomic finality with XRPL.

Finding 2: Multilateral Netting delivers Liquidity Savings, with Cycle Length governing the Latency–Liquidity Trade-off

The DNS simulation achieves an LSR of 82.66%, closely matching the 82% BoF-PSS2 benchmark (Leinonen and Soramäki 2003). The resulting liquidity requirement of €1,315,335 represents a 42.49% reduction relative to RTGS, translating into a 42.49% reduction in total settlement cost — the two percentages are identical because at baseline $r = 3\%$ the delay penalty is negligible, meaning $\text{TCS} \approx L \times r$ and cost reductions are proportional to liquidity reductions. The maximum SF1–SF3 gap under DNS is 2 hours by construction,

with an average gap of 0.8881 hours. The hybrid configuration applies the same multilateral-netting mechanism at a shorter 1-hour cycle, achieving a smaller liquidity reduction of 25.1% ($L^{\text{hybrid}} = \text{€}1,712,511$) but cutting the average settlement gap to 0.3781 hours — approximately half that of DNS. This contrast makes the latency–liquidity trade-off internal to netting architectures explicit: halving the cycle length recovers roughly half the liquidity saving and roughly halves the settlement gap, anchoring the hybrid as the interior point of the frontier reported in Section 5.2.

Finding 3: XRPL Liquidity Requirements under the two concepts

Under the gross-capital concept, the XRPL pool requirement coincides with RTGS prefunding ($MD \times P_{\text{XRP}} = \text{€}2,287,349 \approx \text{€}2,287,265$). Under the risk-adjusted volatility-buffer concept,

$$L_{\text{buffer}}^{\text{XRPL}} = MD \times \sigma_{\text{XRP}} \times P_{\text{XRP}} = 1,971,853 \times 0.05 \times 1.16 = \text{€}114,367 \quad (8)$$

a 95% reduction relative to RTGS. Whether XRPL achieves a structural liquidity advantage therefore depends entirely on which concept is adopted: the gross-capital reading shows none, while the buffer reading shows a large one. Total slippage costs of €80 are second-order under either concept.

Finding 4: Settlement Rates and Operational Reliability

All five configurations achieve high settlement rates under baseline conditions. Under RTGS, 99.80% of payments settle, with 32 payments (0.20%) remaining unsettled due to liquidity constraints, consistent with TARGET2 operational benchmarks (European Central Bank 2010). Under DNS and the hybrid, all payments settle by construction at the end of each clearing cycle (2 hours for DNS, 1 hour for the hybrid). Under wCBDC, the settlement rate is effectively 100% because atomic transfer of central bank money on a permissioned ledger entails no rollback mechanism. Under XRPL, the settlement rate of 99.9874% reflects the stylised $p_{\text{rollback}} = 10^{-4}$ per transaction (Section 3.3), with average latency of 0.001112 hours (approximately 4 seconds).

Finding 4b: Decomposing the Atomic-Settlement and Public-Blockchain effects

The wCBDC control separates the two effects. Costed on gross capital, wCBDC attains a TCS of €68,620 — essentially identical to RTGS (€68,618) — despite delivering atomic finality, near-zero recognition lag, and the same contagion and Herstatt containment as XRPL. Atomic settlement therefore does *not*, by itself, reduce liquidity cost: its benefits are concentrated in latency and systemic-risk containment. The gap between wCBDC and XRPL is correspondingly small under gross capital — the €80 slippage charge, the genuine “public-blockchain” routing cost. The large headline cost advantage of XRPL emerges only under the volatility-buffer specification (€3,511), a difference of €65,109 relative to wCBDC. Since wCBDC possesses identical atomic settlement yet shows no such reduction, this €65,109 is an artefact of the buffer’s risk-adjustment accounting rather than a property of atomic settlement. The robust, framing-independent findings are thus the latency, contagion, and Herstatt advantages, which wCBDC and XRPL share.

5.2 Efficiency Frontier

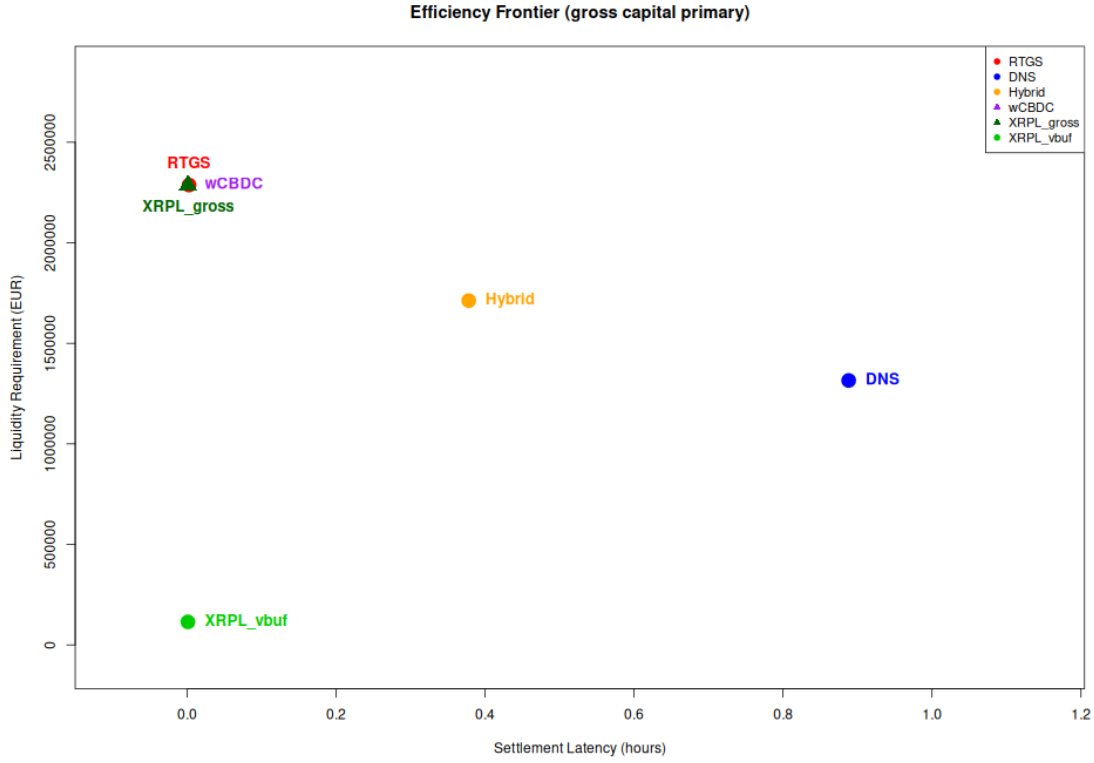


Figure 1: Efficiency Frontier on the Latency–Liquidity Plane ($\lambda = 17,500$, $r = 3\%$, $P_{XRP} = \text{€}1.16$). Five configurations are plotted: RTGS, DNS, the hybrid RTGS–LSM benchmark, wCBDC, and XRPL. XRPL appears at both the gross pool value ($L_{\text{gross}}^{\text{XRPL}} = \text{€}2,287,349$) and the volatility-buffer value ($L_{\text{buffer}}^{\text{XRPL}} = \text{€}114,367$). See Sections 3.2 and 5.1 for the distinction.

Finding 5: Atomic Settlement relocates the Latency Frontier unconditionally and shifts the Liquidity Frontier only under Risk-Adjustment

The configurations occupy the following coordinates on the latency–liquidity plane:

- RTGS: $D = 0.0026$ hrs, $L = \text{€}2,287,265$, $\text{TCS} = \text{€}68,618$
- DNS: $D = 0.8881$ hrs, $L = \text{€}1,315,335$, $\text{TCS} = \text{€}39,460$
- Hybrid: $D = 0.3781$ hrs, $L = \text{€}1,712,511$, $\text{TCS} = \text{€}51,375$
- wCBDC: $D = 0.0011$ hrs, $L = \text{€}2,287,349$, $\text{TCS} = \text{€}68,620$
- XRPL_{Gross}: $D = 0.0011$ hrs, $L = \text{€}2,287,349$, $\text{TCS} = \text{€}68,701$
- XRPL_{Buffer}: $D = 0.0011$ hrs, $L = \text{€}114,367$, $\text{TCS} = \text{€}3,511$

The latency frontier shift is unconditional: XRPL and wCBDC attain the lowest latency (0.0011 hours) regardless of the liquidity concept, far below RTGS and DNS, with the hybrid intermediate. The liquidity frontier shift, by contrast, is conditional. Under the risk-adjusted volatility buffer, XRPL simultaneously attains the lowest latency and the lowest liquidity, satisfying both frontier-shift conditions. Under gross capital, XRPL and wCBDC match RTGS on liquidity and are exceeded by DNS, so atomic settlement relocates along the latency axis rather than shifting the liquidity frontier outright. The structural elimination of the latency–liquidity trade-off claimed by Berndsen (2018) therefore holds in full only under the buffer interpretation; on the latency dimension alone it holds unconditionally.

Finding 6: Identification of r^*

Table 3: Interest Rate Sensitivity under the gross-capital specification.

r	RTGS	DNS	Hybrid	wCBDC	XRPL (gross)	XRPL (buffer)	Winner
0.0%	€0	€0	€0	€0	€80	€80	—
1.0%	€22,873	€13,153	€17,125	€22,873	€22,954	€1,224	DNS
2.0%	€45,745	€26,307	€34,250	€45,747	€45,827	€2,368	DNS
3.0%	€68,618	€39,460	€51,375	€68,620	€68,701	€3,511	DNS
4.0%	€91,491	€52,613	€68,500	€91,494	€91,574	€4,655	DNS
5.0%	€114,363	€65,767	€85,626	€114,367	€114,448	€5,799	DNS
6.0%	€137,236	€78,920	€102,751	€137,241	€137,321	€6,942	DNS
7.0%	€160,109	€92,074	€119,876	€160,114	€160,195	€8,086	DNS
8.0%	€182,981	€105,227	€137,001	€182,988	€183,068	€9,230	DNS

Notes: “Winner” is the lowest-TCS architecture under gross capital — DNS at every positive rate. Under the volatility-buffer specification (final column) XRPL is lowest at every positive rate. The $r = 0\%$ row is economically vacuous: all opportunity costs vanish and only XRPL’s €80 slippage remains.

The existence of a threshold rate r^* is conditional on the liquidity concept. Under the gross-capital specification, DNS attains the lowest total cost at every positive interest rate from 0% to 8%, so no r^* exists at which XRPL becomes the cheapest architecture: its gross liquidity requirement coincides with RTGS and exceeds DNS, and this ordering is invariant in r because the TCS function is approximately linear in r at this calibration. The hybrid configuration sits between RTGS and DNS at every tested rate, exactly as its interior frontier position predicts, and never attains the lowest cost under gross capital. Under the risk-adjusted volatility buffer, by contrast, XRPL attains the lowest total cost at every positive rate, so r^* is effectively zero. The threshold is therefore not a structural constant of the technology but a

function of how on-demand liquidity is costed — the central qualification that the wCBDC control of Finding 4b makes explicit.

Finding 7: Volume Sensitivity and the DNS–Pool Crossover

Table 4: Volume Sensitivity under the gross-capital specification ($r = 3\%$).

λ	RTGS	DNS	Hybrid	wCBDC	XRPL (gross)	XRPL (buffer)	Winner
10,000	€39,748	€24,585	€33,403	€68,620	€68,659	€3,469	DNS
15,000	€60,250	€32,744	€42,909	€68,620	€68,705	€3,516	DNS
20,000	€75,672	€33,471	€45,018	€68,620	€68,677	€3,487	DNS
25,000	€93,212	€35,231	€50,217	€68,620	€68,684	€3,495	DNS
30,000	€117,948	€47,872	€63,833	€68,620	€68,751	€3,561	DNS
35,000	€132,426	€46,914	€62,629	€68,620	€68,706	€3,516	DNS
40,000	€151,559	€48,103	€68,071	€68,620	€68,740	€3,550	DNS
45,000	€170,449	€56,191	€77,919	€68,620	€68,781	€3,592	DNS
50,000	€185,691	€62,773	€81,909	€68,620	€68,806	€3,617	DNS
55,000	€204,633	€56,391	€74,494	€68,620	€68,804	€3,614	DNS
60,000	€222,138	€56,435	€81,288	€68,620	€68,818	€3,628	DNS
65,000	€241,655	€60,555	€89,734	€68,620	€68,836	€3,647	DNS
70,000	€259,962	€63,764	€90,336	€68,620	€68,886	€3,696	DNS
75,000	€277,620	€75,013	€101,050	€68,620	€68,927	€3,738	wCBDC
80,000	€295,729	€69,124	€106,663	€68,620	€68,919	€3,730	wCBDC
85,000	€315,108	€69,009	€100,968	€68,620	€68,894	€3,705	wCBDC
90,000	€334,568	€78,476	€107,661	€68,620	€68,954	€3,765	wCBDC
95,000	€361,622	€74,798	€103,452	€68,620	€68,982	€3,792	wCBDC
100,000	€375,345	€81,578	€117,082	€68,620	€69,110	€3,920	wCBDC

Under gross capital, DNS attains the lowest total cost up to approximately $\lambda = 70,000$ payments per day. Beyond that volume the fixed-pool atomic architectures overtake DNS: wCBDC and XRPL hold a liquidity requirement that is independent of volume, whereas DNS net funding grows with the payment population, so the two cross at $\lambda \approx 75,000$. No λ^* exists in the opposite direction — the atomic-pool architectures never *lose* their relative position as volume rises, since RTGS liquidity scales approximately linearly (a $9.4\times$ increase in TCS for a $10\times$ rise in volume) while the pool requirement is constant. Under the risk-adjusted volatility buffer XRPL is lowest at every tested volume, with its TCS rising only 13.0% across the range (from €3,469 to €3,920). The volume result is therefore that the atomic-pool advantage strengthens monotonically with volume, but whether it dominates DNS at a given volume depends on the liquidity concept. A sustained increase in XRP price is the more plausible threat to buffer-concept cost competitiveness: the breakeven price $P_{\text{XRP}}^* \approx \text{€}13.34$

is approximately 11.5 times the March 2026 price of €1.16 and more than four times XRP's all-time high of €3.11, placing it outside any historically observed price regime.

5.3 Stress Testing and Contagion Dynamics

Table 5: Contagion Analysis: Hub Default Stress Test ($\lambda = 50,000$, 1,000 Monte Carlo runs)

Metric	RTGS	DNS	Hybrid	wCBDC	XRPL
Mean EAF	0.5011	0.0463	0.0308	0.000174 ^a	0.000174 ^a
Affected nodes	49	24	24	49 ^b	49 ^b
Avg loss per node	€45,570	€8,422	€5,627	€16	€16
95th pct EAF	0.5259	0.0697	0.0468	0.000174 ^a	0.000174 ^a
Monte Carlo SE	0.0004	0.0004	0.0003	0.000000 ^a	0.000000 ^a
Relative SE	0.08%	0.86%	0.97%	0.00%	0.00%

^a Analytically derived as $5/28,800$ (maximum finality window / total operating seconds); zero Monte Carlo variance by construction. wCBDC shares XRPL's atomic finality and therefore the same in-flight fraction, isolating the contagion-containment property as an atomic-settlement effect rather than a public-blockchain one.

^b High affected node count for atomic architectures reflects hub connectivity (degree 47/49); average loss per node of €16 confirms near-zero exposure despite broad reach.

Finding 8: Concentrated vs. Diffuse Contagion

The five configurations span three contagion regimes. RTGS generates high-magnitude, full-breadth contagion (49 nodes, mean loss €45,570 per node) without ex ante containment. The netting family compresses in two modes: DNS (2-hour cycle) reduces breadth to 24 net creditors at €8,422 per node, and the hybrid (1-hour cycle) affects the same 24 nodes but cuts the per-node loss to €5,627, with mean EAF 0.0308 against DNS's 0.0463. Shorter cycles accumulate fewer payments before default, so residual net positions are smaller in absolute terms. The higher netting frequency therefore involves a trade-off: it reduces the hybrid's liquidity efficiency (Finding 2) but tightens its contagion containment. The atomic architectures (XRPL and wCBDC) operate on magnitude instead: they affect 49 nodes by hub connectivity (Finding 9) but cut the per-node loss to €16, reflecting the near-complete elimination of secondary exposure through atomic finality. The choice of architecture therefore governs both the magnitude and the distribution of systemic losses. That wCBDC matches XRPL on every contagion metric identifies containment as an atomic-settlement property, not a public-blockchain one.

Finding 9: Atomic Finality reduces EAF by construction

The mean EAF of 0.000174, attained identically by XRPL and wCBDC, corresponds to a 99.97% reduction relative to RTGS and 99.62% relative to DNS, but this figure is analytically determined rather than stochastically estimated. It equals the ratio of the maximum finality window to total operating seconds,

$$f_{\text{in-flight}} = \frac{5 \text{ seconds}}{32 \times 0.25 \times 3,600 \text{ seconds}} = \frac{5}{28,800} \approx 0.000174, \quad (9)$$

exact by construction and producing zero Monte Carlo variance by definition rather than by convergence. wCBDC reproduces this identity exactly because it shares atomic finality with XRPL while differing on every public-blockchain dimension; the matching EAF is the contagion-side mirror of Finding 4b, attributing containment to atomic settlement rather than to the permissionless ledger.

The substantive interpretation is correspondingly narrower than the headline reduction suggests. Reducing settlement gaps to seconds mechanically reduces in-flight exposure in the same proportion, eliminating the recognition lag identified by Berndsen (2018), but the simulation does not stress-test contagion: validator coordination failure, market maker withdrawal under stress, and rollback above the static $p_{\text{rollback}} = 10^{-4}$ baseline are not modelled (Section 6.5). These caveats apply to XRPL alone; wCBDC's reproduction of the same EAF is delivered under permissioned central-bank governance with no validator or market maker dependencies, so its contagion bound is conditional only on atomic central-bank settlement working as designed.

What the cross-architecture comparison does isolate is the decoupling between breadth and magnitude. The 49 affected nodes under atomic settlement reflect hub connectivity (degree 47/49) rather than meaningful exposure: with 5 seconds of in-flight risk, the average loss per node is €16. The same connectivity translates into €45,570 per node under RTGS; €8,422 across 24 net creditors under DNS (2-hour netting); €5,627 across the same 24 nodes under the hybrid (1-hour netting). The progression — full magnitude under gross settlement, compressed under netting, near-zero under atomic finality — is the structural property the

EAF metric is designed to surface.

Finding 10: Monte Carlo robustness

Across 1,000 independent Monte Carlo runs, the standard errors on all EAF estimates are below 0.001 in absolute terms (Table 5), which corresponds to less than 1% of each architecture’s mean EAF, well within the 5% threshold for relative standard error. The qualitative ranking of the five settlement mechanisms is preserved in every one of the 1,000 runs, confirming that the contagion ordering established in Finding 9 is not an artefact of any particular random draw.

5.4 Cross-Border Settlement

Table 6: Herstatt Risk: Correspondent Banking, CLS, XRPL, and wCBDC ($P = \text{€}1,000,000$, 100,000 MC runs)

Metric	Correspondent	CLS	XRPL	wCBDC
Settlement gap Δt	24 hrs	5 hrs	0 hrs	0 hrs
Analytical E[Loss]	€2,397	€500	€0	€0
MC default rate	0.24%	0.06%	0.00%	0.00%
MC unconditional E[Loss]	€2,360	€550	€0	€0
MC conditional E[Loss default]	€1,000,000	€1,000,000	N/A	N/A
Risk reduction vs correspondent	—	79.1%	100%	100%
Default rate SE	0.0153%	0.0074%	0%	0%

Finding 11: XRPL eliminates Herstatt Risk

XRPL achieves 100% Herstatt risk elimination through zero settlement gap atomic PvP, and the wCBDC control achieves the same elimination for the same structural reason, confirming that the result is a property of atomic payment-versus-payment settlement rather than of the public-blockchain implementation. CLS reduces risk through frequency reduction — default rate falls from 0.24% to 0.06% — but not severity mitigation: conditional losses under both CLS and correspondent banking equal the full transaction principal of €1,000,000. The distinction between unconditional and conditional expected loss is therefore critical: the unconditional expected loss averages across all simulated transactions including those where no

default occurs, reflecting both probability and severity simultaneously (€2,360 for correspondent banking, €550 for CLS). The conditional expected loss isolates severity by conditioning on default events only — and here CLS offers no improvement over correspondent banking, since when a default does occur within the settlement gap, the entire principal of €1,000,000 is lost regardless of whether the gap is 24 hours or 5 hours. CLS therefore reduces Herstatt risk through exposure frequency reduction, while XRPL eliminates it structurally by reducing the settlement gap to zero. The analytical expected loss of €2,397 under correspondent banking is derived directly from the formula $\mathbb{E}[\text{Loss}] = P \cdot (1 - (1 - p)^{\Delta t})$, while the MC unconditional expected loss of €2,360 reflects sampling variation across 100,000 runs — the small difference confirms Monte Carlo convergence to the analytical benchmark.

Finding 12: Trapped Liquidity and MD^*

Table 7: Trapped Liquidity: Correspondent Banking, CLS, XRPL, and wCBDC (1,000 daily FX transactions, $r = 3\%$)

Metric	Correspondent	CLS	XRPL	wCBDC
Gross pool / nostro requirement	€200,000,000	€60,000,000	€2,287,349	€2,287,349
Volatility buffer	—	—	€114,367	—
Opportunity cost (on-demand basis)	€6,000,000	€1,800,000	€3,431	€68,620
Opportunity cost (gross-pool basis)	—	—	€68,620 ^a	—
Opportunity cost saving vs correspondent	—	€4,200,000	€5,996,569	€5,931,380
Opportunity cost (gross-pool) vs correspondent	—	—	€5,931,380 ^a	—
Gross pool reduction vs correspondent	—	70.0%	98.9%	98.9%

Notes: XRPL and wCBDC share the gross corridor pool by construction, so the 98.9% gross-pool reduction is an atomic-settlement effect common to both architectures. ^a Two opportunity-cost bases are reported for XRPL: the on-demand basis charges the 3% rate only on the volatility buffer; the gross-pool basis treats the entire gross pool as encumbered idle capital. Under the latter, XRPL coincides with wCBDC by construction.

XRPL reduces gross corridor liquidity requirements by 98.9% relative to correspondent banking, from €200,000,000 in nostro prefunding to a gross pool requirement of €2,287,349 — a reduction driven by atomic settlement and therefore shared with wCBDC (Table 7). The resulting opportunity cost depends on how the corridor pool is treated. Under the on-demand basis, only the volatility buffer $L_{\text{buffer}}^{\text{XRPL}} = €114,367$ — the capital actually at risk from XRP price movements during the 3–5 second settlement window — is committed, giving an opportunity cost of €3,431, a 99.9% reduction against the €6,000,000 cost of correspondent nostro balances. Under the conservative gross-pool basis, the entire €2,287,349 pool is

treated as encumbered, giving €68,620 (98.9% reduction) and coinciding with wCBDC by construction.

The pool depth threshold MD^* is defined as the point at which the gross XRPL pool value in EUR equals the nostro prefunding requirement — the breakeven condition at which the two architectures have identical liquidity costs. Setting $MD^* \times P_{\text{XRP}} = \text{Nostro prefunding}$ and solving gives:

$$MD^* = \frac{\text{Nostro prefunding}}{P_{\text{XRP}}} = \frac{€200,000,000}{€1.16} = 172,413,793 \text{ XRP} \equiv €200,000,000 \quad (10)$$

where the equivalence $\equiv €200,000,000$ confirms internal consistency: converting MD^* back to EUR at the current exchange rate recovers the nostro prefunding amount by construction. The current gross pool of $MD = 1,971,853 \text{ XRP}$ is 87.5 times smaller than MD^* , meaning the corridor pool would need to grow by a factor of 87.5 before XRPL loses its liquidity cost advantage relative to correspondent nostro prefunding. This provides substantial headroom for institutional adoption and volume growth, confirming that XRPL's cost advantage is robust even under significant expansion of on-demand liquidity corridors.

An analogous breakeven threshold can be constructed against CLS, the principal alternative cross-border architecture for Herstatt-eligible currency corridors. MD_{CLS}^* is the XRPL pool depth (in XRP) at which the gross XRPL pool value in EUR equals CLS's prefunding requirement; CLS itself operates in central bank money and holds no XRP. Applying the same formula with the €60,000,000 CLS prefunding baseline (Table 7) gives:

$$MD_{\text{CLS}}^* = \frac{\text{CLS prefunding}}{P_{\text{XRP}}} = \frac{€60,000,000}{€1.16} = 51,724,138 \text{ XRP} \equiv €60,000,000. \quad (11)$$

The XRP figure denotes how large the XRPL pool would have to be to match the CLS funding scale, not anything CLS itself holds. The current gross pool of $MD = 1,971,853 \text{ XRP}$ is 26.2 times smaller than MD_{CLS}^* , so XRPL retains a corridor-liquidity advantage against CLS until its pool grows by a factor of 26.2. The smaller margin relative to correspondent banking is the expected consequence of CLS's multilateral payment-versus-payment netting, which

already compresses funding requirements by approximately 70% relative to raw correspondent banking (Table 7). Two structural caveats accompany this comparison. First, CLS coverage is restricted to 18 eligible currencies, whereas correspondent banking applies universally; the correspondent-banking threshold therefore remains the binding constraint for the majority of global cross-border corridors. Second, CLS funding is centralised at CLS Bank to support multilateral netting, whereas nostro prefunding is distributed per-corridor — the MD^* concept maps more cleanly onto the latter as a corridor-level liquidity commitment. Both thresholds nonetheless support the same qualitative finding: XRPL’s cost advantage is robust under any plausible expansion of on-demand liquidity corridors, against either incumbent benchmark.

5.5 Sensitivity Analysis

Table 8: Sensitivity Analysis Summary: lowest-cost architecture by dimension

Parameter	Range	Gross-capital winner	Buffer winner
r	[0%, 8%] (9)	DNS (8) [†]	XRPL (8) [†]
λ	[10,000, 100,000] (19)	DNS (13), wCBDC (6)	XRPL (19)
α	{0.05, 0.1, 0.2} (3)	DNS (3)	XRPL (3)
β	{1.2, 1.8, 2.5} (3)	DNS (3)	XRPL (3)
κ	{0.03, ..., 0.07} (5)	DNS (5)	XRPL (5)
δ	{1.0, 1.5, 2.0} (3)	DNS (3)	XRPL (3)
γ	{2.0, 2.5, 3.0} (3)	DNS (3)	XRPL (3)
Total	45	DNS 38 / wCBDC 6	XRPL 44

Note: [†]The $r = 0\%$ cell has no meaningful winner under either concept: all opportunity costs vanish and only XRPL’s €80 slippage remains, making XRPL the most expensive rather than the cheapest. The cell is excluded from the per-concept totals on both sides. The latency, contagion, and Herstatt orderings (not shown) hold in 45/45 cells under both concepts.

The key message is twofold. First, under gross capital DNS is the lowest-cost architecture in 38 of 45 cells and the fixed-pool atomic architectures (wCBDC, XRPL) in the remaining 6 at high volume and positive interest rates; XRPL never attains the lowest cost under gross capital. Under the volatility buffer XRPL is lowest in 44 of 45 cells, with $r = 0\%$ excluded as vacuous. Second, and independently of the liquidity concept, the qualitative latency, contagion, and Herstatt orderings are invariant across all 45 cells: atomic settlement attains the lowest latency, the lowest EAF, and full Herstatt elimination in every parameterisation. The α , β , κ , and δ dimensions leave the cost ordering unchanged because delay penalties and slippage are

second-order at this calibration; the γ sweep confirms independence from the power-law exponent, with the hub-to-spoke ratio stable between 23.5 and 24.0. The robustness that matters for the thesis is therefore structural: the latency and resilience advantages of atomic settlement hold across every tested parameterisation, while the cost ranking is governed by the liquidity concept rather than by any single parameter.

6 Research Limitations

While the proposed methodology provides a structured and rigorous framework for analysing settlement systems, several limitations must be acknowledged when interpreting the results.

6.1 Model Scope and Network Scale

The simulation network comprises 50 nodes (5 hubs, 45 spokes), representing a stylised approximation of real-world large-value payment systems. TARGET2 operates with approximately 1,000 direct participants (European Central Bank 2024), meaning the simulated network is approximately 20 times smaller than the actual infrastructure. Absolute EUR values should not be interpreted as estimates of real-world settlement costs; what is meaningful is the relative ordering of costs and the parametric conditions under which this ordering changes. Furthermore, while this study assumes linear scaling of liquidity requirements, it is established in the literature that larger payment networks often exhibit non-linear efficiencies in liquidity recycling. In a full-scale system like TARGET2, the higher probability of offsetting payment flows may increase the Liquidity Saving Ratio (LSR) for traditional architectures (Leinonen and Soramäki 2005), potentially widening the cost advantage of DNS and the hybrid relative to the fixed-pool atomic architectures under the gross-capital concept. This reinforces the conclusion that the cost case for atomic settlement rests on the volatility-buffer interpretation rather than on a structural liquidity advantage.

The hub-to-spoke degree ratio of 23.5 is consistent with empirical interbank network topology (Soramäki et al. 2007; Boss et al. 2004), but formal estimation of γ is unreliable at $n = 50$. Sensitivity analysis over $\gamma \in \{2.0, 2.5, 3.0\}$ confirms robustness. The `recycle_delay`

parameter of 0.421 was calibrated under a coarser 8-step discretisation (one hour per step) by minimising the squared deviation between the simulated gridlock probability and the TARGET2 benchmark of approximately 15% under 3x stress volume, producing a simulated gridlock probability of 14.80%. The parameter is carried forward to the 32-step framework because re-calibration at this granularity is infeasible: per-step payment volume is too thin for gridlock to manifest as a simulation observable, making the calibration target unobservable at 15-minute step intervals. The limitation is therefore one of transferability across discretisations rather than arbitrary parameter choice — the value is empirically grounded but its validity at finer granularity cannot be confirmed from simulation evidence alone. Its direction of bias is conservative for the central findings: a higher recycling delay reduces intraday liquidity availability, which would increase RTGS queuing and TCS, only strengthening the XRPL cost advantage. Sensitivity to this parameter is limited in practice: since the RTGS delay penalty $\varphi(D^{\text{RTGS}}) = \text{€}0.00$ at baseline, varying `recycle_delay` across its plausible range affects queuing dynamics but not total settlement cost materially, leaving the cost ordering unchanged. The peak-to-average liquidity ratio of 1.36x remains below the TARGET2 benchmark of 3x (European Central Bank 2010), reflecting the absence of strategic payment timing behaviour discussed in Section 6.3, and represents a lower bound on intraday liquidity concentration — meaning RTGS TCS is understated relative to a fully calibrated model.

6.2 Data Availability and Synthetic Inputs

Due to the confidential nature of large-value payment system data, the simulation is calibrated using aggregate statistics from the BIS and ECB (Bank for International Settlements 2020; European Central Bank 2010). While $\kappa = 0.05$ is derived to be consistent with the Hernandez Cruz et al. (2024) observed slippage range at institutional-scale pool depth, as described in Section 4.2, a scale mismatch remains between the payment sizes at which that range was observed and the synthetic payment population of this simulation: the observed slippage range of 1.36%–1.73% corresponds to trades of approximately €955,675, which is 6,414 times the median synthetic payment of €149. At the synthetic median payment size, the slippage formula produces negligible cost per transaction; at hub-to-hub large payments in the right tail

of the log-normal distribution, slippage may be understated relative to the Hernandez Cruz et al. (2024) regime. The net direction of bias on aggregate Σ_{slip} is ambiguous: the majority of payments are small (understating slippage per transaction) while the few large payments drive most of the aggregate cost (potentially overstating it at observed institutional scale). Since $\Sigma_{\text{slip}} = \text{€}80$ represents only 2.28% of total XRPL TCS, even a substantial rescaling of slippage costs would not affect the qualitative cost ordering.

A related concern is pool depth under tail-risk conditions. On a stress day with large hub-to-hub payments in the upper tail of the distribution, the ratio V/MD may exceed the range over which the slippage formula is empirically grounded, potentially producing underestimates of market impact for individual large transactions. The simulation does not model order book depletion or market maker withdrawal under stress — both of which would increase effective slippage non-linearly on high-volume days. This represents a genuine limitation of the XRPL cost estimates under tail-risk scenarios: while $\Sigma_{\text{slip}} = \text{€}80$ is negligible at baseline, slippage costs could be substantially higher on days with concentrated large-value flows, and this uncertainty is not fully captured by the κ and δ sensitivity analysis. Addressing this would require order-book depth data from institutional-scale XRP trading venues, which is not currently available in the academic literature.

The XRP/EUR exchange rate $P_{XRP} = \text{€}1.16$ is verified against CoinGecko data as of March 28, 2026 (CoinGecko 2026). A 10% price increase scales both liquidity concepts linearly: under the volatility buffer, XRPL TCS rises from $\text{€}3,511$ to approximately $\text{€}3,854$, leaving XRPL the cheapest architecture; under gross capital, the pool value rises from $\text{€}2,287,349$ to $\text{€}2,516,084$ and XRPL TCS from $\text{€}68,701$ to approximately $\text{€}75,563$, slightly exceeding RTGS ($\text{€}68,618$) but still well above DNS ($\text{€}39,460$). The qualitative cost ranking is unaffected on either concept: DNS remains the cheapest under gross capital and XRPL remains the cheapest under the buffer. The baseline XRP daily volatility of $\sigma_{XRP} = 5\%$ is a conservative assumption relative to realised market conditions. Using daily closing prices from CoinGecko over the two-year window from April 20, 2024 to April 20, 2026 (CoinGecko 2026), the sample standard deviation of 730 daily returns yields a realised daily volatility of 4.46% (price data and calculation in the code repository; see Section 8), corresponding to an

annualised volatility of 85.19% ($\sigma_{\text{daily}} \times \sqrt{365}$). The adopted baseline of 5% therefore lies above the two-year realised average, meaning the reported XRPL liquidity requirement $L_{\text{buffer}}^{\text{XRPL}}$ is overstated relative to recent market conditions. Since $L_{\text{buffer}}^{\text{XRPL}}$ scales linearly with σ_{XRP} , any reduction toward the realised average would further widen the cost advantage relative to both RTGS and DNS. At 10% volatility — more than twice the realised average — XRPL TCS on the on-demand basis (volatility buffer only) would rise to approximately €6,942, still substantially below DNS (€39,460), confirming that the qualitative cost ordering is robust across the full plausible range of XRP volatility estimates.

Similarly, the baseline opportunity cost of capital $r = 3\%$ is set slightly above the prevailing ECB deposit facility rate of 2.00% (as of the calibration date), so as not to understate the cost of liquidity locked in settlement. As with the volatility assumption, this is a conservative choice applied symmetrically across all five architectures, and no qualitative result depends on it: the rate sensitivity analysis of Section 5.2 confirms the cost ordering is invariant over $r \in [0\%, 8\%]$, including at the prevailing 2% level.

Considering the same basis, a related sensitivity concerns XRP price appreciation. The breakeven price $P_{\text{XRP}}^* \approx \text{€}13.34$ — the price at which XRPL TCS under the volatility-buffer specification rises to equal DNS TCS, assuming all other parameters remain unchanged — is approximately 11.5 times the March 2026 price of €1.16 and more than four times XRP's all-time high of approximately €3.11 reached in January 2018. This places the breakeven condition outside any historically observed price regime, making it implausible rather than a tail risk under current market conditions. However, the assumption that all other parameters remain unchanged is unlikely to hold in practice: a sustained price increase of this magnitude would simultaneously alter other parameters in offsetting directions. Higher prices attract deeper market making activity, increasing MD and partially offsetting the rise in $L_{\text{buffer}}^{\text{XRPL}}$. Historically, XRP price spikes have also been accompanied by elevated volatility, meaning σ_{XRP} would likely rise alongside P_{XRP} , amplifying the cost increase beyond the linear price effect. The net impact on XRPL TCS under a sustained price appreciation scenario is therefore ambiguous and cannot be determined from the static simulation framework adopted here — a limitation that would require a dynamic model with endogenous liquidity

provision to resolve properly, identified as a direction for future research in Section 7.3. Furthermore, sustained institutional adoption — the primary mechanism through which P_{XRP} could approach the breakeven level — would simultaneously deepen ODL corridors, improve pathfinding efficiency, and reduce per-transaction costs through network effects, partially or fully offsetting the price appreciation effect on $L_{\text{buffer}}^{\text{XRPL}}$ through the same adoption dynamic that drives the price increase.

The pool depth parameter MD presents a particular data availability challenge specific to the XRPL architecture. In Ripple’s On-Demand Liquidity product, cross-border payments are executed by purchasing XRP on a source exchange, transferring it across the XRP Ledger, and selling it on a destination exchange. The effective liquidity available for a given payment therefore depends on the combined order book depth of two separate exchanges at opposite ends of the payment corridor, which varies by corridor, exchange, time of day, payment size, and prevailing market maker activity. No single published figure for ODL corridor depth exists in the academic literature or in Ripple’s public disclosures, and corridor-specific estimates would not generalise across the stylised network modelled in this simulation. The MD parameter is therefore calibrated internally to maintain proportional consistency with the 50-node network scale, as described in Section 4.2, in the same way all other liquidity parameters are scaled to the stylised network rather than to absolute real-world magnitudes. Sensitivity analysis over $\kappa \in \{0.03, 0.04, 0.05, 0.06, 0.07\}$ and $\delta \in \{1.0, 1.5, 2.0\}$ confirms that the qualitative cost ordering is robust to uncertainty in both the slippage parameters and the pool depth calibration.

6.3 Model Abstraction

The simulation does not model intraday liquidity recycling, which may overstate the RTGS liquidity requirement (Manning et al. 2009). Relatedly, the model characterises steady-state settlement and does not capture the liquidity dynamics specific to the opening and closing phases of the RTGS day, which run on distinct mechanics (Section 4.1). Participant behaviour follows predefined rules, excluding strategic responses under stress documented by Bech and Garratt (2003). In particular, strategic liquidity hoarding — whereby participants withhold

outgoing payments upon observing stress, as documented by Bech and Garratt (2003) — would amplify RTGS contagion beyond the EAF values reported in Table 5, since withheld inflows reduce hub liquidity and cascade outward through the network. Under XRPL atomic settlement, no such channel exists: each payment settles or fails independently within 3–5 seconds, eliminating the queuing mechanism through which hoarding propagates. The absence of strategic behaviour in the simulation therefore represents a conservative assumption for the XRPL contagion results. The hybrid configuration similarly abstracts from the strategic payment-submission timing on which LSM effectiveness depends in practice (Bech and Garratt 2003), and should be read as a stylised representation of the multilateral-offsetting mechanism rather than a full behavioural model.

6.4 Settlement Finality Scope

The analysis assumes functional equivalence between operational and legal finality for XRPL, while recognising that legal harmonisation with the EU Settlement Finality Directive (Bank for International Settlements 2012) remains an open institutional question. Operationally, XRPL achieves deterministic finality within 3–5 seconds. Legally, however, settlement finality requires recognition under applicable insolvency law — without which a court could potentially unwind a settled transaction in the event of counterparty insolvency, reintroducing the principal risk the system was designed to eliminate. This does not negate the liquidity and latency advantages reported in Tables 2 and 5, which are operational properties independent of legal classification. It does, however, mean that the full elimination of settlement risk reported in Section 5.4 is conditional on legal recognition being in place — a prerequisite that currently varies across jurisdictions and is identified as a policy priority in Section 7.2. The legal recognition of XRPL settlement finality is conceptually distinct from the broader regulatory classification of XRP as an asset class: the resolution of *SEC v. Ripple Labs* (U.S. Securities and Exchange Commission 2025) — comprising the district court’s final judgment of 7 August 2024 and the dismissal of cross-appeals in 2025 — substantially clarified the latter under US securities law but did not establish settlement-finality recognition under EU insolvency law, which remains the binding constraint on institutional adoption identified in

Section 7.2.

6.5 Scope of Contagion Analysis

The Section 4.5 finding that XRPL atomic settlement reduces the in-flight exposure fraction to $5/28,800$ holds by construction, conditional on the consensus, liquidity, and finality assumptions of the simulation framework being satisfied in the stress state. Three mechanisms specific to blockchain-based settlement are not modelled within the framework and accordingly bound the strength of the conclusion.

First, validator coordination failure. XRPL finality relies on the Unique Node List delivering supermajority agreement within the 3–5 second window; coordinated validator failure, compromise, or governance dispute during a stress event would suspend or delay finality at the moment the contagion-containment property is most needed. The simulation treats finality as deterministic at the stylised baseline $p_{\text{rollback}} = 10^{-4}$, which captures consensus failure as a static baseline but not as a state-dependent stress variable; the formal analysis of Chase and MacBrough (2018) shows that the protocol’s safety and liveness are in fact conditional on UNL overlap and can fail under adverse network conditions — precisely the stress states not modelled here.

Second, market maker withdrawal under stress. As noted in Section 6.2 in the context of slippage, the simulation does not model order-book depletion or market maker withdrawal. The contagion implication is sharper than the slippage implication: if market makers withdraw on-demand liquidity during a hub-default event, affected payments cease to settle rather than settling at higher cost, and the in-flight exposure window for queued payments expands beyond the 3–5 second finality bound assumed in the EAF derivation. The Section 4.5 EAF therefore describes the in-flight fraction conditional on continuous market maker provision, not unconditional on stress-state liquidity supply.

Third, rollback probability elevation. The $p_{\text{rollback}} = 10^{-4}$ baseline is set to represent ordinary operating conditions, not to stress states in which validator coordination, network connectivity, or transaction-flow concentration could plausibly push consensus performance above this

baseline. No stress-conditional rollback function is implemented, and the result is conditional on the static baseline holding through the hub-default scenario.

These three mechanisms collectively define the boundary of the Section 4.5 finding: the simulation verifies the arithmetic of EAF reduction under atomic finality; it does not test the assumptions under which that arithmetic obtains. Resolving the limitation would require either an extended framework with stress-state-conditional finality, validator, and liquidity functions, or empirical observation of XRPL behaviour under documented institutional-scale stress events. The scope limit reinforces the policy framing of Section 7.2: XRPL adoption involves a reallocation of risk from the liquidity and latency dimensions of traditional architectures to the consensus, governance, and corridor-liquidity dimensions of blockchain-based settlement, rather than the wholesale elimination of systemic risk.

7 Conclusion and Policy Implications

7.1 Summary of Findings

This thesis set out to investigate the economic conditions under which blockchain-based settlement mechanisms can improve efficiency and risk allocation relative to traditional clearing and settlement infrastructures. Five principal conclusions emerge from the simulation-based analysis.

Conclusion 1: XRPL's total-cost ranking depends on how its liquidity is costed. Under the conservative gross-capital specification, DNS attains the lowest total settlement cost and XRPL is comparable to RTGS; XRPL's headline cost advantage (TCS €3,511 at $r = 3\%$, $\lambda = 17,500$) emerges only under the risk-adjusted volatility-buffer specification. The threshold rate r^* is therefore not a structural constant but a function of the liquidity concept: effectively zero under the buffer, and never reached under gross capital.

Conclusion 2: XRPL relocates the latency frontier unconditionally and shifts the liquidity frontier only under risk-adjustment. XRPL and wCBDC attain the lowest latency (0.0011 hours) under either liquidity concept. A simultaneous liquidity and latency frontier shift holds

only under the volatility buffer; under gross capital XRPL matches RTGS on liquidity and is exceeded by DNS, so atomic settlement relocates along the latency axis rather than shifting the liquidity frontier outright.

Conclusion 3: XRPL reduces contagion by three to four orders of magnitude. Mean EAF is 0.5011 (RTGS), 0.0463 (DNS), and 0.000174 (XRPL) across 1,000 Monte Carlo runs. A novel finding of concentrated versus diffuse contagion under RTGS and DNS respectively has direct regulatory implications for the distribution of systemic losses.

Conclusion 4: Atomic PvP eliminates Herstatt risk and reduces trapped liquidity. Monte Carlo simulation across 100,000 runs confirms 100% Herstatt risk elimination for both atomic architectures, XRPL and wCBDC. The gross corridor pool value required for either (€2,287,349) is 87.5 times lower than correspondent nostro prefunding (€200,000,000) — an atomic-settlement effect shared by both configurations and unconditional on the liquidity concept. The further reduction to an opportunity cost of €3,431 is specific to XRPL under the volatility-buffer interpretation (Section 3.2); under the symmetric gross-capital treatment, wCBDC carries the full €68,620 opportunity cost of its CB-money corridor pool, with the €65,109 gap between the two attributable to buffer accounting rather than to atomic settlement (Finding 4b). $MD^* = €200,000,000$ provides substantial headroom above current pool requirements, confirming that the gross corridor-liquidity advantage of atomic settlement is robust to significant growth in institutional payment volumes regardless of which atomic implementation is adopted.

Conclusion 5: The robust findings are the latency and resilience advantages, which are framing-independent; the cost ranking is not. Across 45 univariate sensitivity cells, the latency, contagion, and Herstatt orderings hold in every case under both liquidity concepts. The total-cost ranking, however, is concept-dependent: under gross capital DNS is cheapest in 38 of 45 cells and the fixed-pool atomic architectures in 6 (high volume and positive interest rate), while under the buffer XRPL is cheapest in 44 of 45. Robustness to validator coordination, market maker withdrawal under stress, and rollback elevation is not tested within the simulation framework and is discussed in Section 6.5.

7.2 Policy Implications

The policy implications below identify institutional vehicles in which the simulation findings can be empirically tested. Where those vehicles are wholesale CBDC pilots, the simulation results transfer because the operational properties driving the cost ordering of Section 5 (atomic finality, low liquidity requirement, near-zero recognition lag) are architecture-agnostic features of atomic-settlement infrastructure rather than properties specific to permissionless XRPL. Wholesale CBDC and XRPL are alternative implementations of the same underlying mechanism the simulation models; the choice between them turns on the governance and risk allocation discussed in Section 3.3 rather than on the operational cost properties measured in Section 5.

Implication 1: Atomic Settlement, not Public Blockchain, drives the operational gains

The wCBDC decomposition implies that the latency and resilience benefits identified in this thesis can be obtained under direct central bank governance, without the validator-concentration and governance risks of a permissionless ledger. The cost case for atomic settlement, by contrast, is conditional on the treatment of on-demand liquidity and does not survive a symmetric gross-capital comparison. The ECB and BIS CPPI should therefore evaluate atomic-settlement adoption primarily as a latency- and resilience-driven proposition, with the cost case treated as contingent on the accounting of corridor capital. Concretely, the Eurosystem's exploratory work on DLT-based wholesale settlement (European Central Bank 2025b) — pairing the Bundesbank-led trigger solution with the longer-term Eurosystem DLT platform (European Central Bank 2025a) — provides an institutional vehicle in which the latency and resilience differentials identified here can be benchmarked against TARGET2 under live operating conditions, and in which the liquidity-cost question can be resolved with observed corridor data.

Implication 2: Atomic Settlement reduces Systemic Risk

The three-to-four order-of-magnitude EAF reduction suggests atomic settlement is not merely an efficiency tool but a systemic stability mechanism. Regulators implementing DORA

and the BIS Principles for FMIs (Bank for International Settlements 2012) should incorporate atomic settlement finality as a resilience criterion. BIS Project Mariana (Bank for International Settlements 2023a), which demonstrated atomic foreign-exchange settlement via wholesale CBDCs across the Banque de France, Monetary Authority of Singapore, and Swiss National Bank, provides the closest existing institutional analogue against which the contagion-containment property modelled in Section 4.5 can be observed at supervisory scale.

Implication 3: On-Demand Liquidity reduces Trapped Capital

The 98.9% reduction in trapped gross corridor liquidity relative to correspondent banking is delivered by both atomic architectures — XRPL and the wCBDC control alike — and is therefore an atomic-settlement effect rather than a public-blockchain one (Finding 4b). $MD^* = \text{€}200,000,000$ provides a concrete quantitative criterion for institutional adoption decisions, directly addressing the policy objectives set out in the G20 Roadmap for Enhancing Cross-border Payments (Financial Stability Board 2020), supported by the systemic scale documented in the CPMI Red Book (Bank for International Settlements 2020). The further reduction in opportunity cost to €3,431 is specific to XRPL under the risk-adjusted volatility-buffer interpretation and is conditional on that accounting choice; under the symmetric gross-capital treatment, both XRPL and wCBDC sit at an opportunity cost of approximately €68,620 on the same gross pool. BIS Project Agorá (Bank for International Settlements 2024), which tests tokenised cross-border settlement across seven central bank jurisdictions, operationalises the gross-pool reduction in a multilateral wCBDC setting directly comparable to the cross-border simulation framework of Section 5.4, providing the empirical context in which the MD^* threshold derived here can be tested against observed corridor depth.

Implication 4: Legal Harmonisation is a Prerequisite

Legal recognition of XRPL settlement finality under the EU Settlement Finality Directive and governance standards for the XRPL Unique Node List must be developed before institutional adoption can be contemplated (Bank for International Settlements 2012; Shin 2026). The EU DLT Pilot Regime (European Union 2022) offers an existing legal sandbox

under which DLT-based market infrastructures can operate with temporary derogations from existing requirements, providing a credible vehicle for staged deployment in which Settlement Finality Directive (European Union 1998) recognition and validator governance standards can be developed iteratively under supervisory observation (European Securities and Markets Authority 2025) rather than through ex ante harmonisation across member states. More broadly, institutional adoption of XRPL requires the development of supervisory frameworks that address validator governance, operational resilience standards, and corridor liquidity requirements — risk dimensions that are absent from traditional FMI oversight regimes and for which regulatory guidance does not yet exist.

7.3 Directions for Future Research

Five directions emerge for future research. First, expanding network scale ($n \geq 500$) to enable reliable γ estimation and TARGET2 benchmark calibration. Second, incorporating game-theoretic behavioural responses consistent with Bech and Garratt (2003). Third, refining slippage parameters against institutional-scale XRPL DEX data. Fourth, extending the legal analysis to encompass the full requirements of the EU Settlement Finality Directive. Fifth, extending the contagion framework to include stress-state-conditional finality, validator coordination failure, and market maker withdrawal mechanisms, as identified in Section 6.5.

7.4 Concluding Remarks

This thesis has examined, through simulation-based analysis, whether blockchain-based atomic settlement via the XRP Ledger can improve efficiency and risk allocation relative to traditional clearing and settlement infrastructures. The findings are best read as conditional where they concern cost and robust where they concern latency and resilience. On the cost dimension, the ranking of architectures depends on how on-demand liquidity is costed: under a conservative gross-capital treatment, deferred net settlement remains the most cost-efficient architecture and atomic settlement offers no liquidity-cost advantage, whereas under a risk-adjusted volatility-buffer treatment XRPL attains the lowest cost. The permissioned wholesale CBDC control makes this dependence precise: an architecture with identical

atomic settlement but no public-blockchain features shows no cost reduction under gross capital, locating XRPL's headline advantage in the buffer accounting rather than in atomic settlement itself. On the latency and resilience dimensions, by contrast, the results are framing-independent and structural. Atomic settlement — whether implemented as a permissionless ledger or a permissioned wholesale CBDC — attains the lowest settlement latency, reduces the Exposure Amplification Factor by three to four orders of magnitude, and eliminates Herstatt risk, in every tested parameterisation and under both liquidity concepts. These are the genuine contributions of atomic settlement to payment-system design, and they hold within the structure of the stylised 50-node model, whose absolute euro values are indicators of relative ordering rather than estimates of real-world cost. The policy reading that follows is measured: institutional adoption of atomic settlement is best motivated by its latency and systemic-resilience properties, which are real and robust, rather than by a cost case contingent on the accounting treatment of corridor liquidity — a question the wholesale-CBDC pilots identified in Section 7.2 are well placed to resolve empirically. Adoption, in any implementation, represents a reallocation of risk from the liquidity and latency dimensions of traditional architectures to the consensus, governance, and corridor-liquidity dimensions of atomic settlement, rather than the wholesale elimination of systemic risk.

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8 R Simulation Code

The complete simulation framework is implemented across seven R scripts and one supporting data file, all available in the accompanying code repository. The simulation is fully reproducible from the random seeds documented in each script.

- `01_network.R` — Network construction and validation
- `02_rtgs.R` — RTGS baseline simulation
- `03_dns.R` — DNS baseline simulation
- `04_xrpl.R` — XRPL atomic settlement (gross and volatility-buffer concepts)
- `05_stress.R` — Stress testing and contagion analysis
- `06_crossborder.R` — Cross-border extension (Herstatt risk, trapped liquidity, MD^*)
- `07_wcbbc.R` — Permissioned wholesale CBDC control and atomic/public-blockchain decomposition
- `07b_hybrid.R` — Hybrid RTGS–LSM benchmark (high-frequency multilateral offsetting)
- `08_sensitivity.R` — Sensitivity analysis and efficiency frontier across all five configurations (run under each liquidity concept)
- `Daily_avg_XRP_volatility_2_years.xlsx` — CoinGecko daily price series (20 April 2024 – 20 April 2026) and realised-volatility calculation supporting the σ_{XRP} calibration of Section 6.2